

The human genome

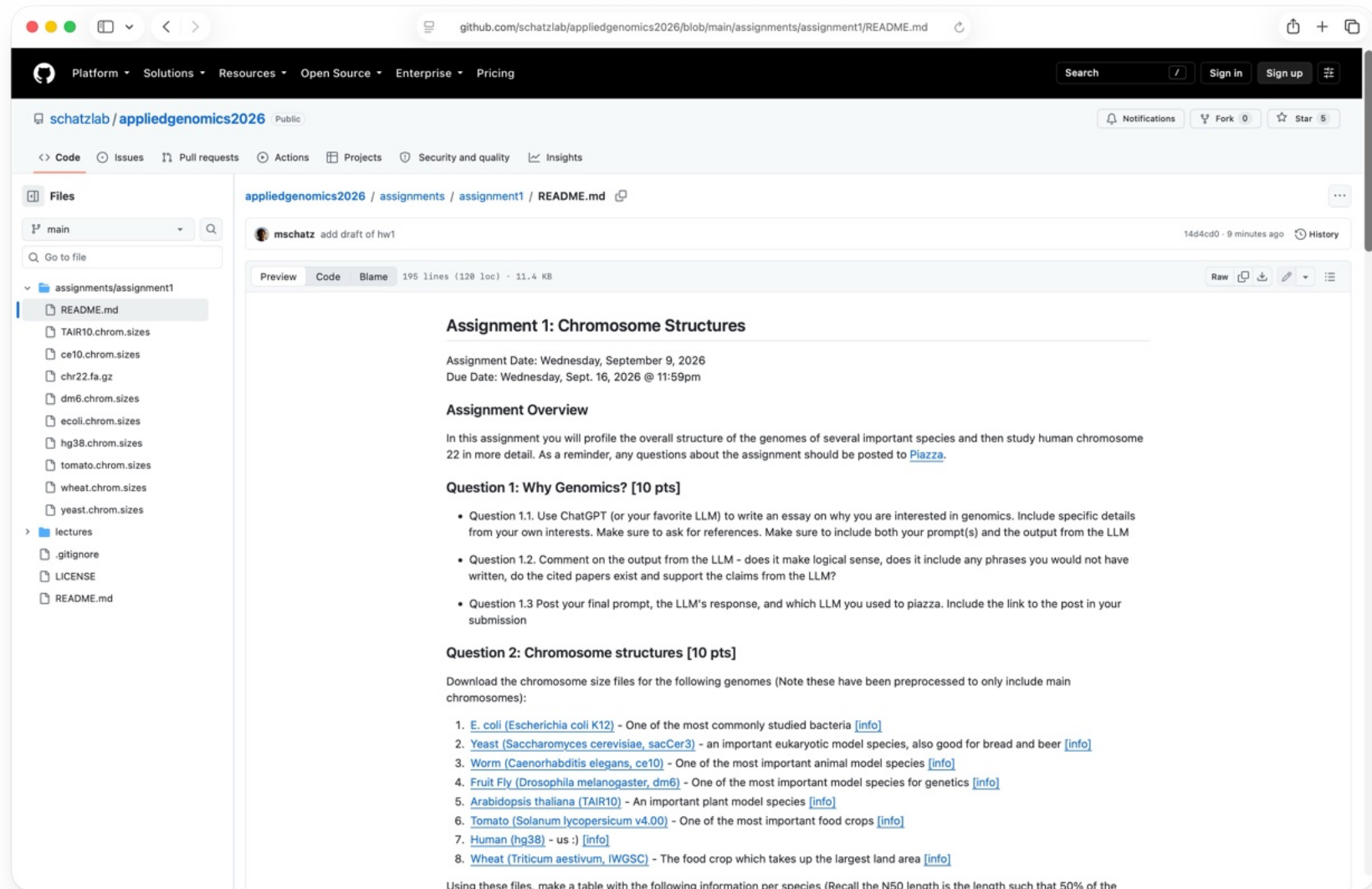
Michael Schatz

Sept 14, 2026

Lecture 4: Applied Comparative Genomics



Assignment I



github.com/schatzlab/appliedgenomics2026/blob/main/assignments/assignment1/README.md

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Files

main

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assignments/assignment1

- README.md
- TAIR10.chrom.sizes
- ce10.chrom.sizes
- chr22.fa.gz
- dm6.chrom.sizes
- ecoli.chrom.sizes
- hg38.chrom.sizes
- tomato.chrom.sizes
- wheat.chrom.sizes
- yeast.chrom.sizes

lectures

- .gitignore
- LICENSE
- README.md

appliedgenomics2026 / assignments / assignment1 / README.md

mschatz add draft of hw1 14d4cd0 · 9 minutes ago History

Preview Code Blame 195 lines (120 loc) · 11.4 KB Raw Download Edit

Assignment 1: Chromosome Structures

Assignment Date: Wednesday, September 9, 2026
Due Date: Wednesday, Sept. 16, 2026 @ 11:59pm

Assignment Overview

In this assignment you will profile the overall structure of the genomes of several important species and then study human chromosome 22 in more detail. As a reminder, any questions about the assignment should be posted to [Piazza](#).

Question 1: Why Genomics? [10 pts]

- Question 1.1. Use ChatGPT (or your favorite LLM) to write an essay on why you are interested in genomics. Include specific details from your own interests. Make sure to ask for references. Make sure to include both your prompt(s) and the output from the LLM
- Question 1.2. Comment on the output from the LLM - does it make logical sense, does it include any phrases you would not have written, do the cited papers exist and support the claims from the LLM?
- Question 1.3 Post your final prompt, the LLM's response, and which LLM you used to piazza. Include the link to the post in your submission

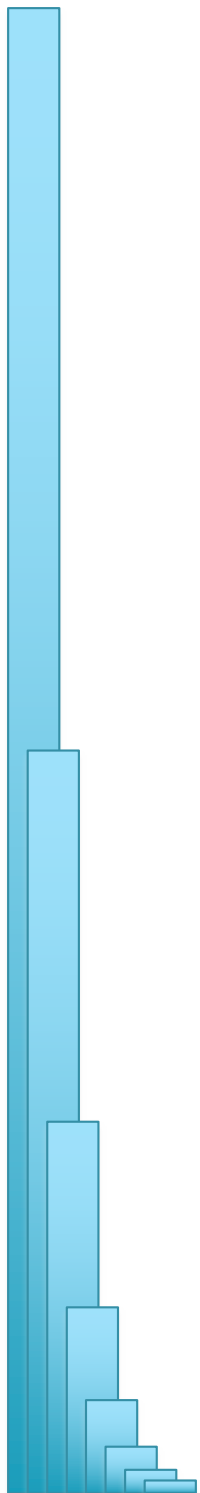
Question 2: Chromosome structures [10 pts]

Download the chromosome size files for the following genomes (Note these have been preprocessed to only include main chromosomes):

- [E. coli \(Escherichia coli K12\)](#) - One of the most commonly studied bacteria [\[info\]](#)
- [Yeast \(Saccharomyces cerevisiae, sacCer3\)](#) - an important eukaryotic model species, also good for bread and beer [\[info\]](#)
- [Worm \(Caenorhabditis elegans, ce10\)](#) - One of the most important animal model species [\[info\]](#)
- [Fruit Fly \(Drosophila melanogaster, dm6\)](#) - One of the most important model species for genetics [\[info\]](#)
- [Arabidopsis thaliana \(TAIR10\)](#) - An important plant model species [\[info\]](#)
- [Tomato \(Solanum lycopersicum v4.00\)](#) - One of the most important food crops [\[info\]](#)
- [Human \(hg38\) - us](#) [\[info\]](#)
- [Wheat \(Triticum aestivum, IWGSC\)](#) - The food crop which takes up the largest land area [\[info\]](#)

Using these files, make a table with the following information per species (Recall the N50 length is the length such that 50% of the

<https://github.com/schatzlab/appliedgenomics2026/tree/main/assignments/assignment1>
Due end of day on Wednesday Sept 16 (right before midnight)



0. Recap

Shredded Book Reconstruction

- Dickens accidentally shreds the first printing of A Tale of Two Cities
 - Text printed on 5 long spools

It was	the best	of times,	it was the	age of wisdom,	it was the	age of foolishness,	...
It was	the best	of times,	it was the	age of wisdom,	it was the	age of foolishness,	...
It was	the best	of times,	it was the	age of wisdom,	it was the	age of foolishness,	...
It was	the best	of times,	it was the	age of wisdom,	it was the	age of foolishness,	...
It	was	the best	of times,	it was the	age of wisdom,	it was the	age of foolishness,

- How can he reconstruct the text?
 - 5 copies x 138,656 words / 5 words per fragment = 138k fragments
 - The short fragments from every copy are mixed together
 - Some fragments are identical

Greedy Reconstruction

It was the best of
age of wisdom, it was
best of times, it was
it was the age of
it was the age of
it was the worst of
of times, it was the
of times, it was the
of wisdom, it was the
the age of wisdom, it
the best of times, it
the worst of times, it
times, it was the age
times, it was the worst
was the age of wisdom,
was the age of foolishness,
was the best of times,
was the worst of times,
wisdom, it was the age
worst of times, it was

It was the best of
was the best of times,
the best of times, it
best of times, it was
of times, it was the
of times, it was the
times, it was the worst
times, it was the age

The repeated sequence make the correct reconstruction ambiguous

- It was the best of times, it was the [worst/age]

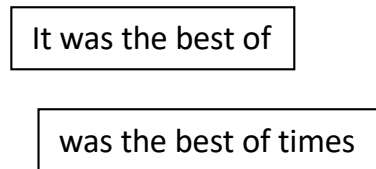
Model the assembly problem as a graph problem

How long will it take to compute the overlaps?

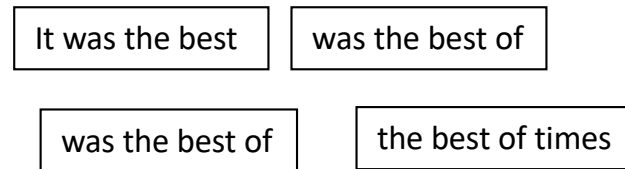
de Bruijn Graph Construction

- $G_k = (V, E)$
 - V = Length- k sub-fragments
 - E = Directed edges between consecutive sub-fragments
 - Sub-fragments overlap by $k-1$ words

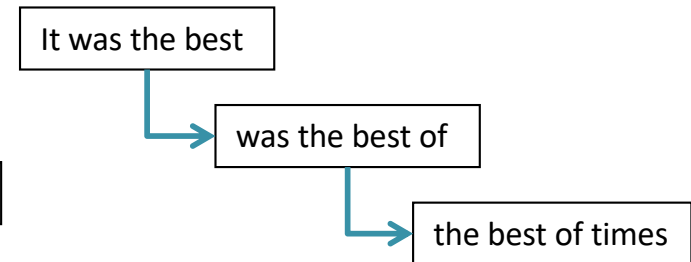
Fragments $|f|=5$



Sub-fragment $k=4$



Directed edges (overlap by $k-1$)



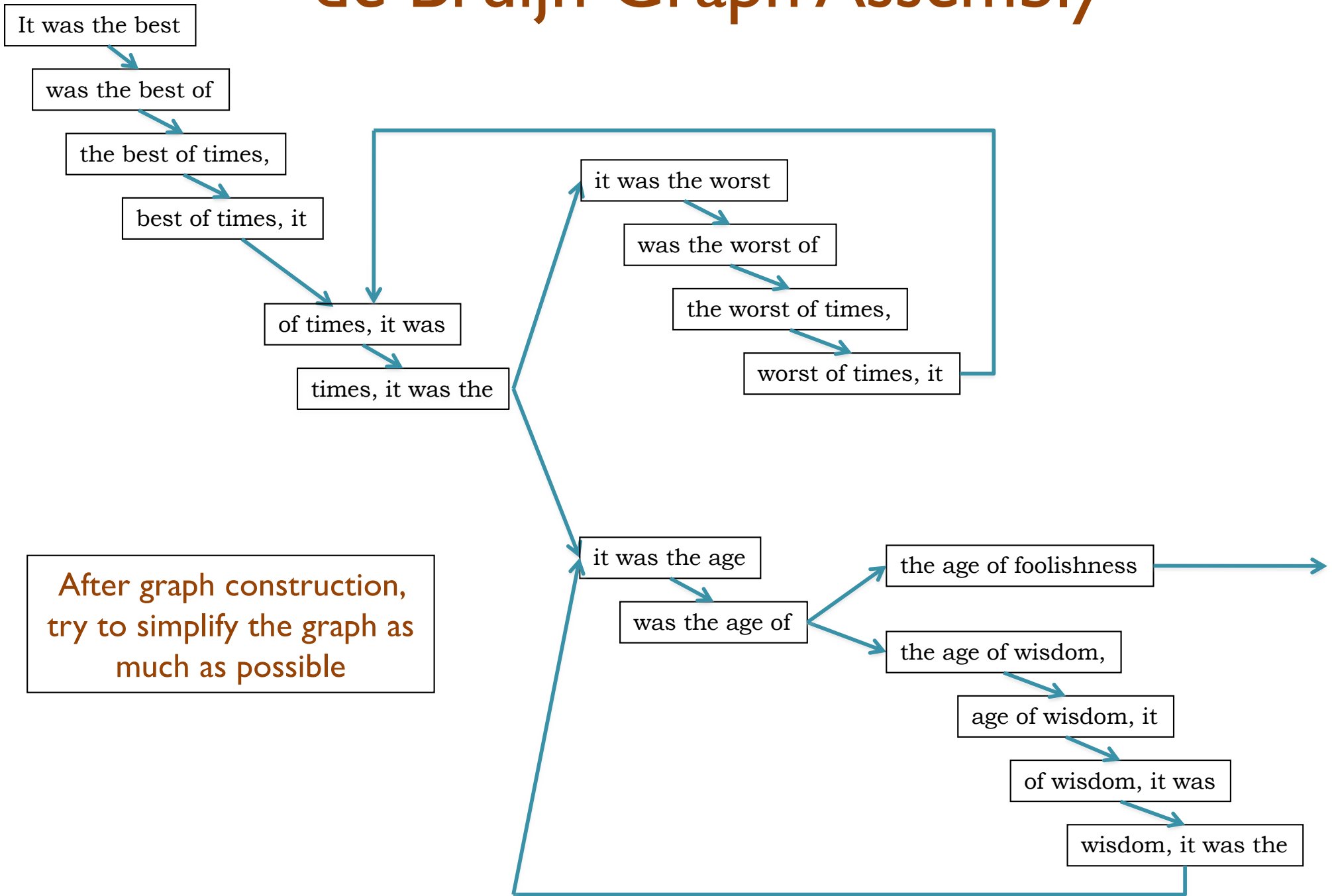
– Overlaps between fragments are implicitly computed

How to pronounce:

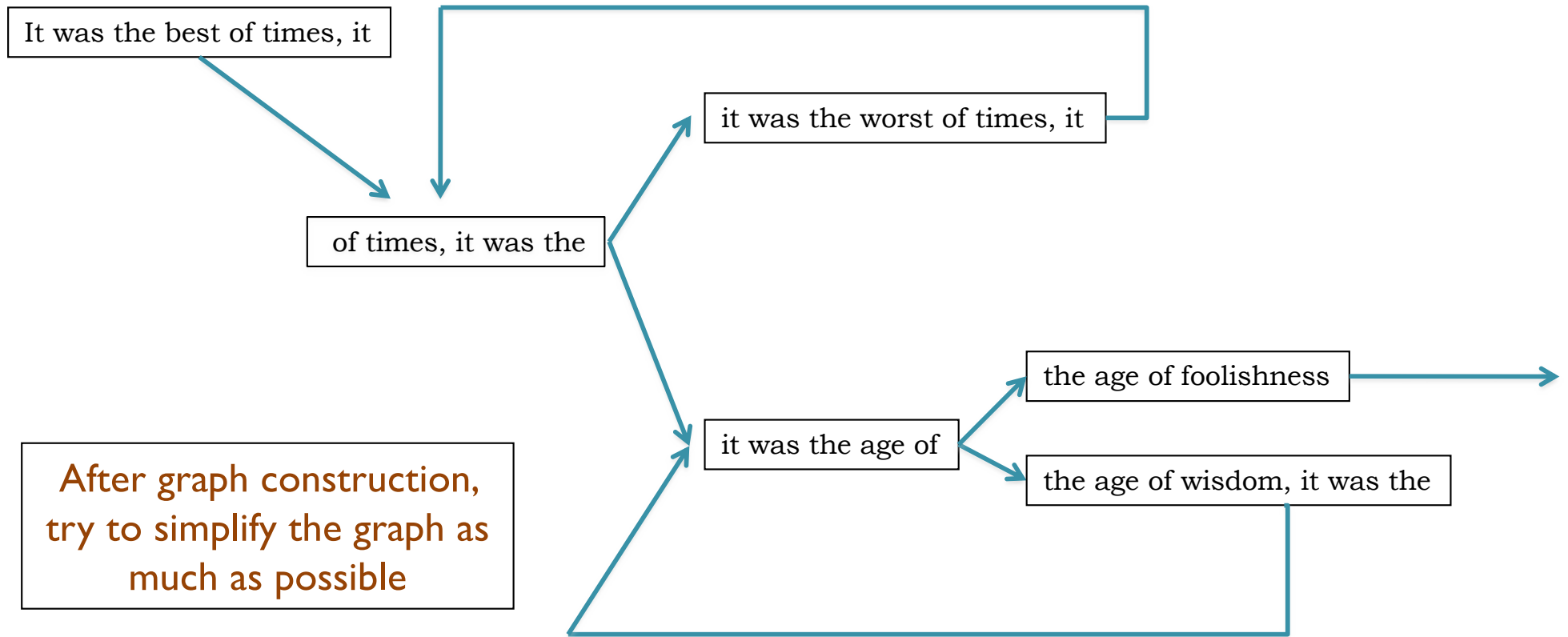
https://forvo.com/word/de_bruijn/

de Bruijn, 1946
Idury et al., 1995
Pevzner et al., 2001

de Bruijn Graph Assembly



de Bruijn Graph Assembly



The full tale

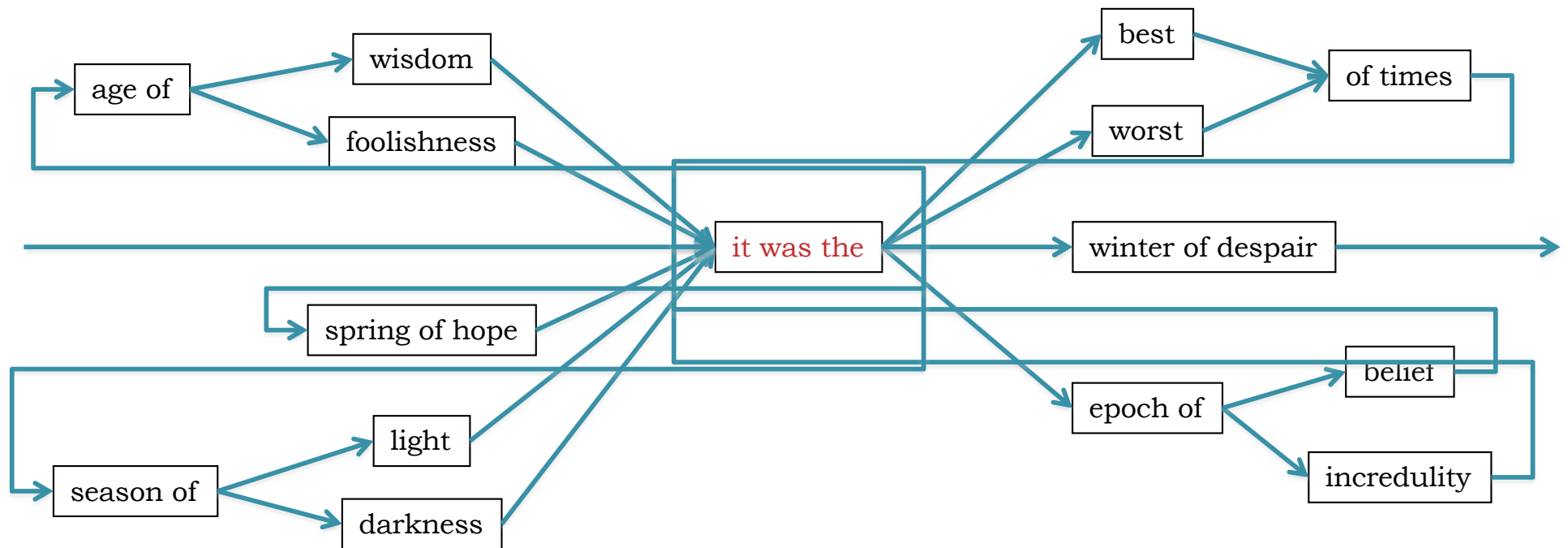
... it was the best of times it was the worst of times ...

... it was the age of wisdom it was the age of foolishness ...

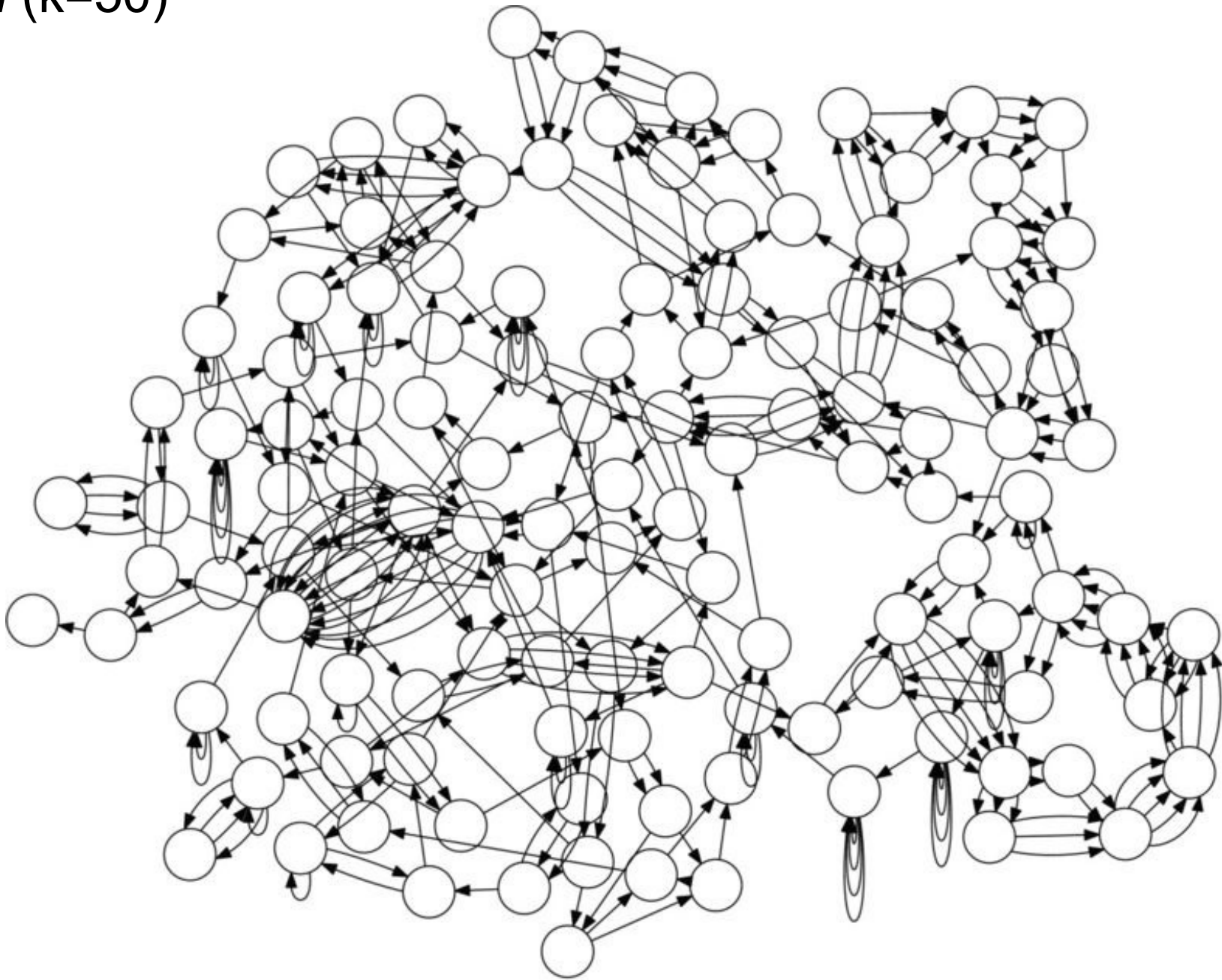
... it was the epoch of belief it was the epoch of incredulity ...

... it was the season of light it was the season of darkness ...

... it was the spring of hope it was the winder of despair ...



E. coli (k=50)



Reducing assembly complexity of microbial genomes with single-molecule sequencing

Koren et al (2013) Genome Biology. **14**:R101 <https://doi.org/10.1186/gb-2013-14-9-r101>

Contig N50

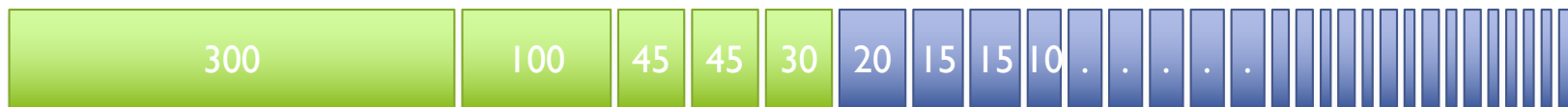
Def: 50% of the genome is in contigs as large as the N50 value

Example: 1 Mbp genome

50%



A



N50 size = 30 kbp

B



N50 size = 3 kbp

Contig N50

Def: 50% of the genome is in contigs as large as the N50 value

50%

Better N50s improves the analysis in every dimension

- Better resolution of genes and flanking regulatory regions
- Better resolution of transposons and other complex sequences
- Better resolution of chromosome organization
- Better sequence for all downstream analysis

Just be careful of N50 inflation!

- *A very very very bad assembler in 1 line of bash:*
- *cat *.reads.fa > genome.fa*

N50 size = 3 kbp

Pop Quiz I

Assemble these reads using a de Bruijn graph approach ($k=3$):

ACACG

ATTAC

GATTA

TTACA

Pop Quiz I

Assemble these reads using a de Bruijn graph approach (k=3):

ACACG : ACA → CAC → ACG
ATTAC : ATT → TTA → TAC
GATTA : GAT → ATT → TTA
TTACA : TTA → TAC → ACA

Pop Quiz I

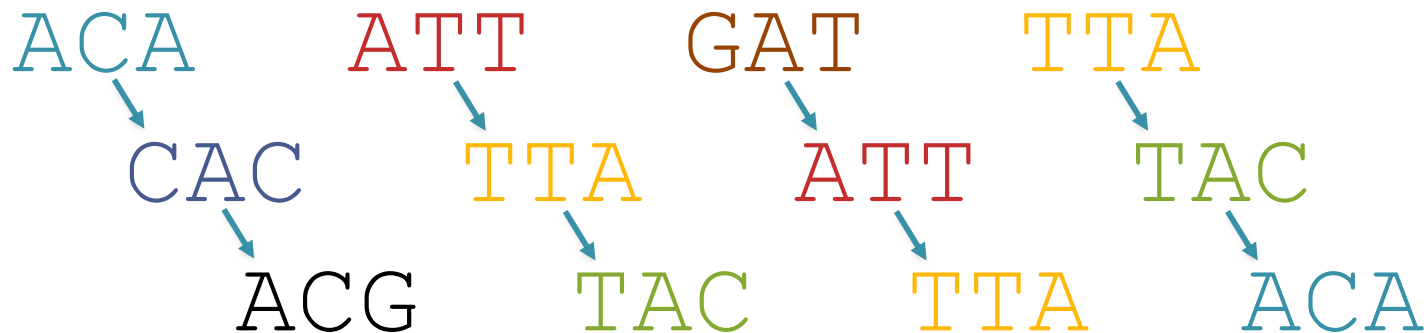
Assemble these reads using a de Bruijn graph approach (k=3):

ACACG: ACA → CAC → ACG

ATTAC: ATT → TTA → TAC

GATTA: GAT → ATT → TTA

TTACA: TTA → TAC → ACA



Pop Quiz I

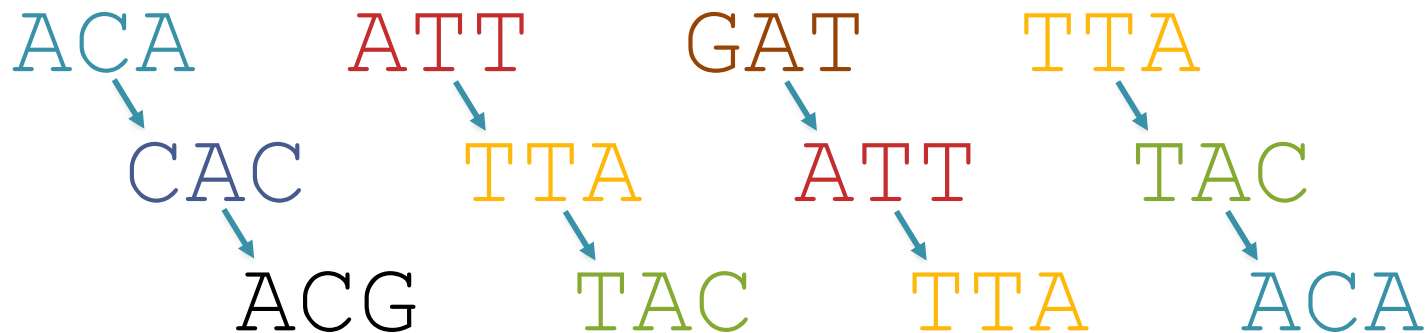
Assemble these reads using a de Bruijn graph approach (k=3):

ACACG

ATTAC

GATTA

TTACA



Pop Quiz I

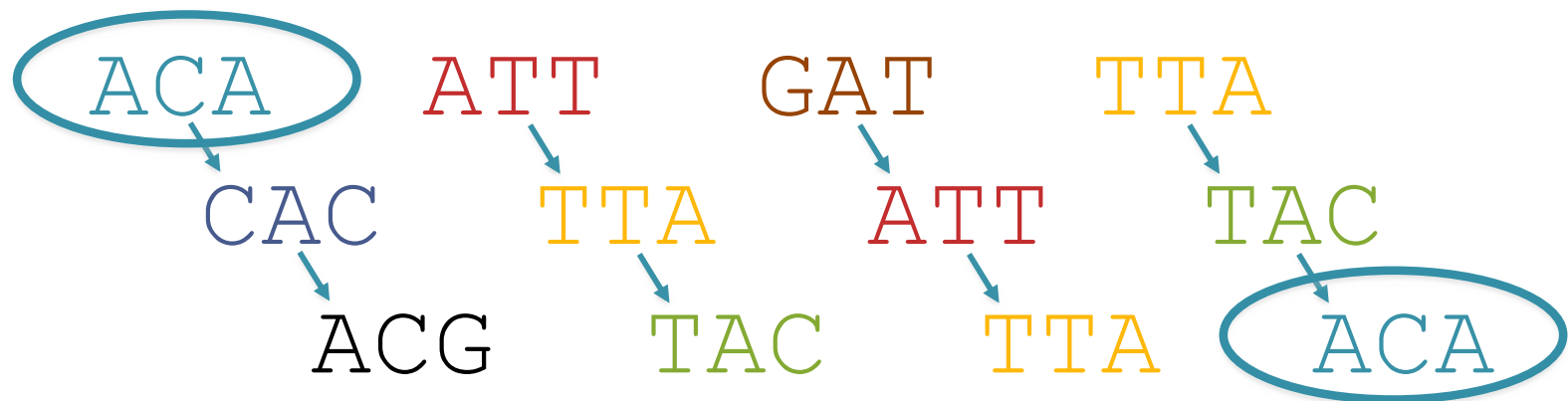
Assemble these reads using a de Bruijn graph approach (k=3):

ACACG

ATTAC

GATTA

TTACA



Pop Quiz I

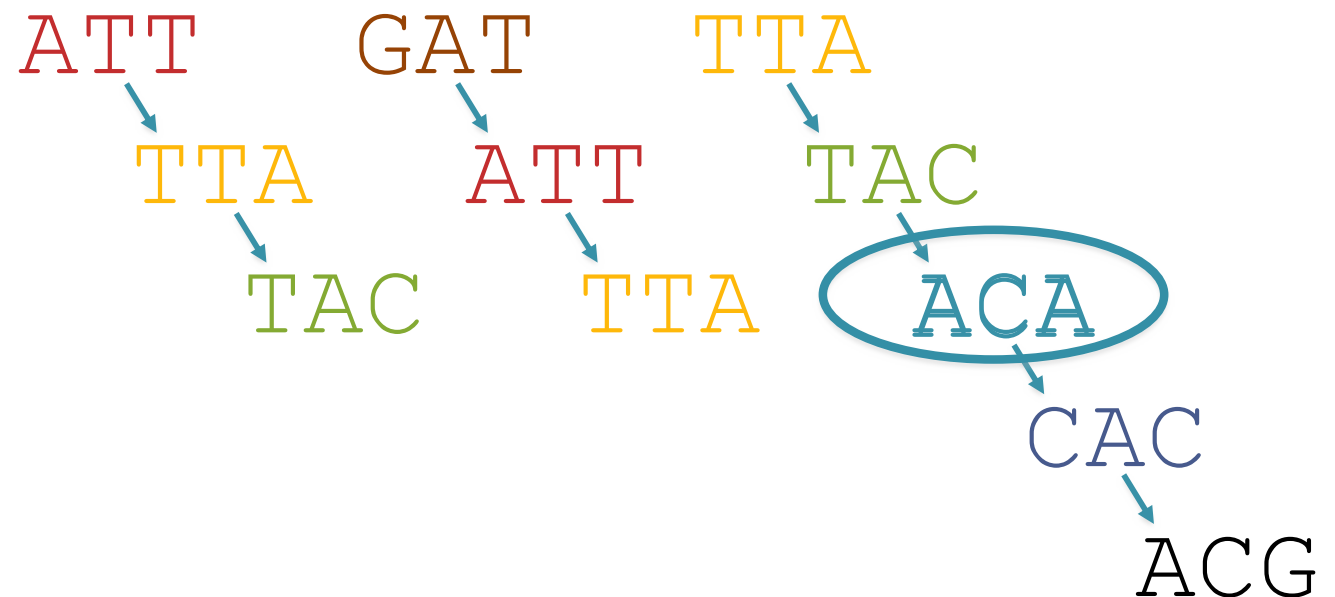
Assemble these reads using a de Bruijn graph approach (k=3):

ACACG

ATTAC

GATTA

TTACA



Pop Quiz I

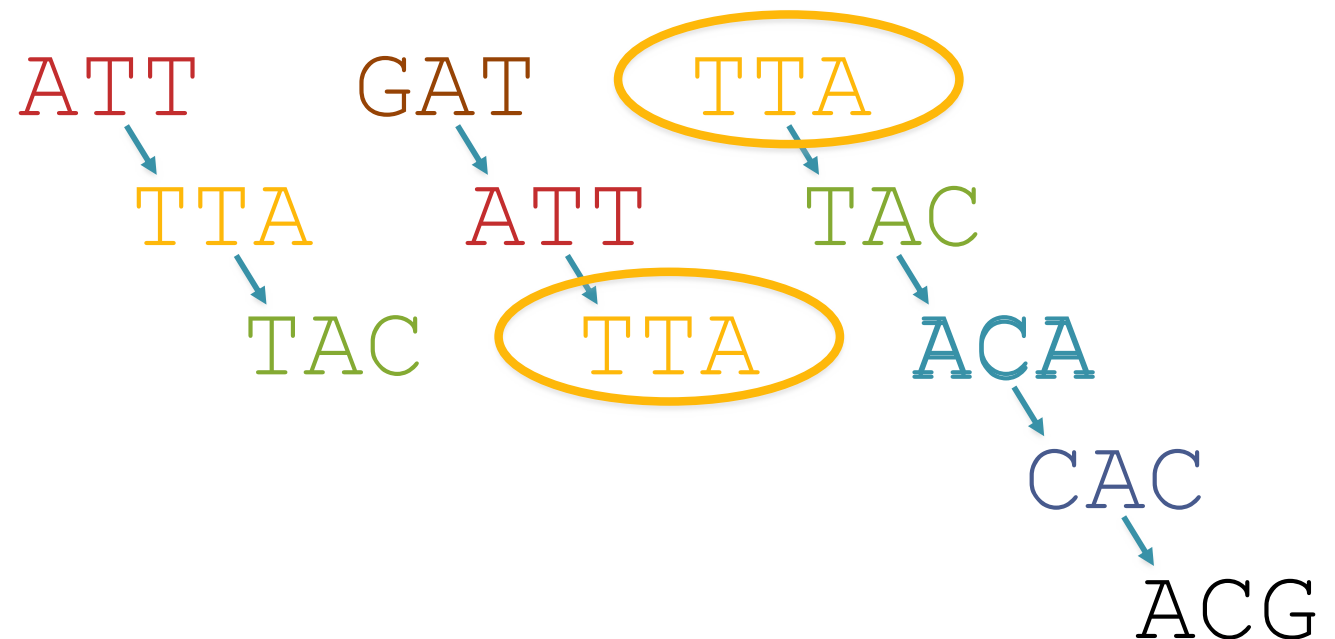
Assemble these reads using a de Bruijn graph approach (k=3):

ACACG

ATTAC

GATTA

TTACA



Pop Quiz I

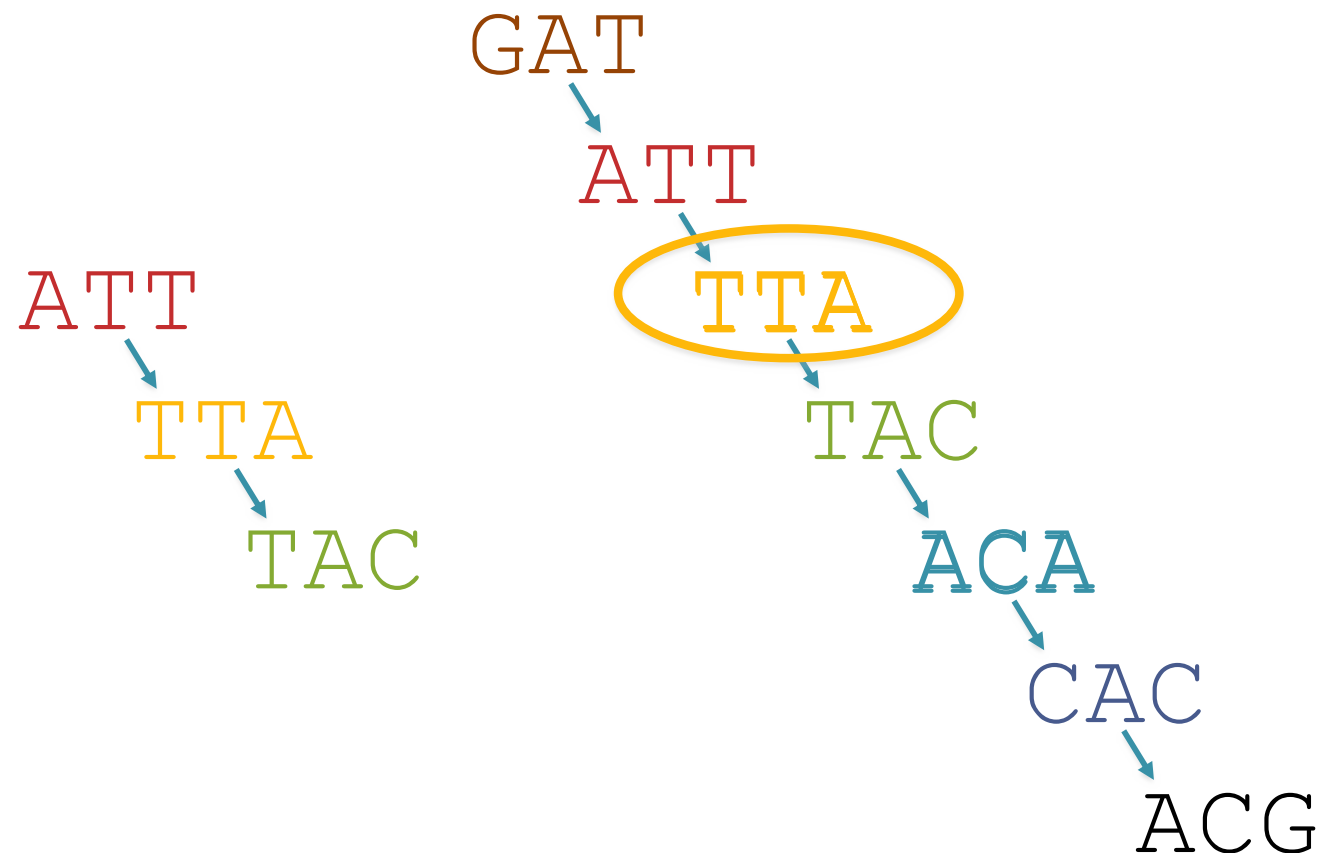
Assemble these reads using a de Bruijn graph approach (k=3):

ACACG

ATTAC

GATTA

TTACA



Pop Quiz I

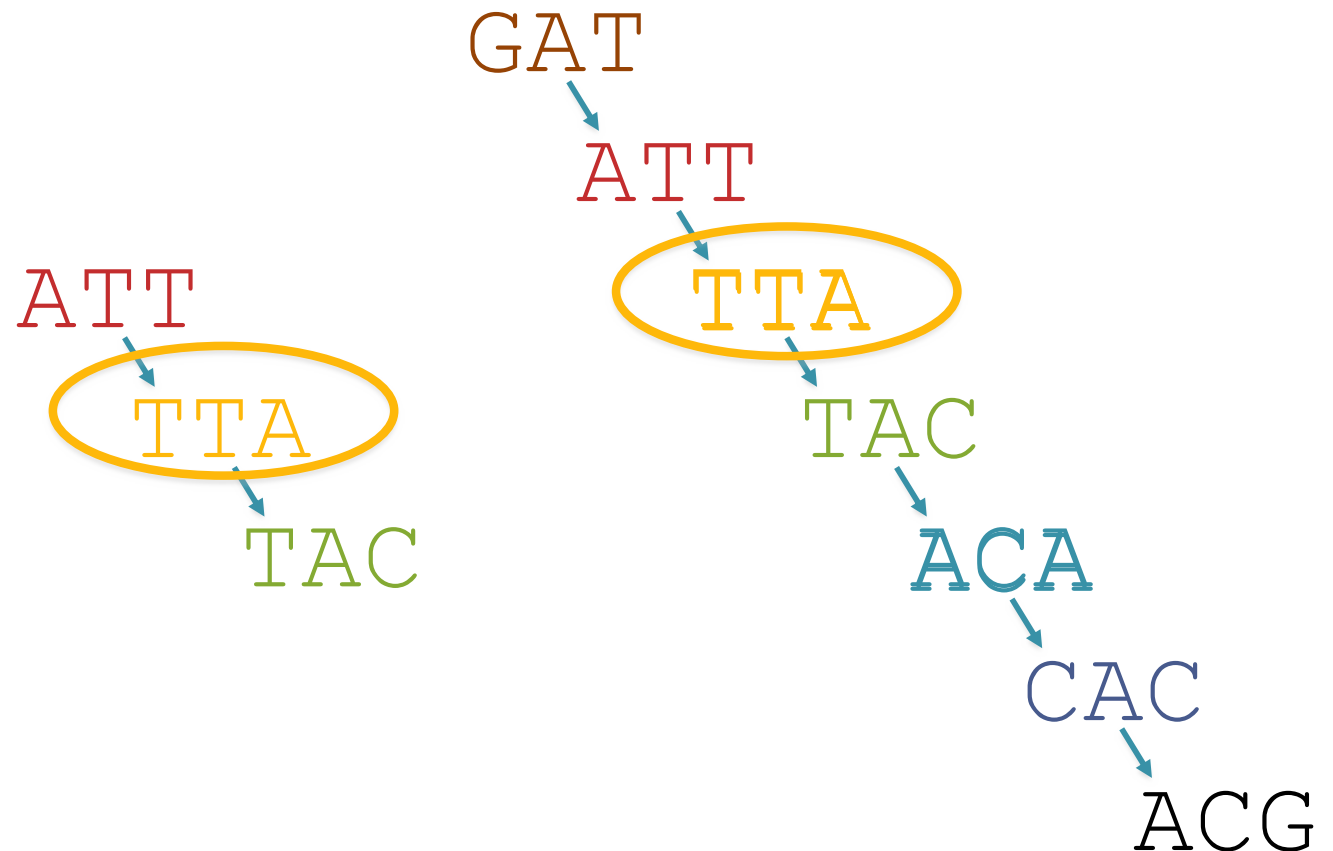
Assemble these reads using a de Bruijn graph approach (k=3):

ACACG

ATTAC

GATTA

TTACA



Pop Quiz I

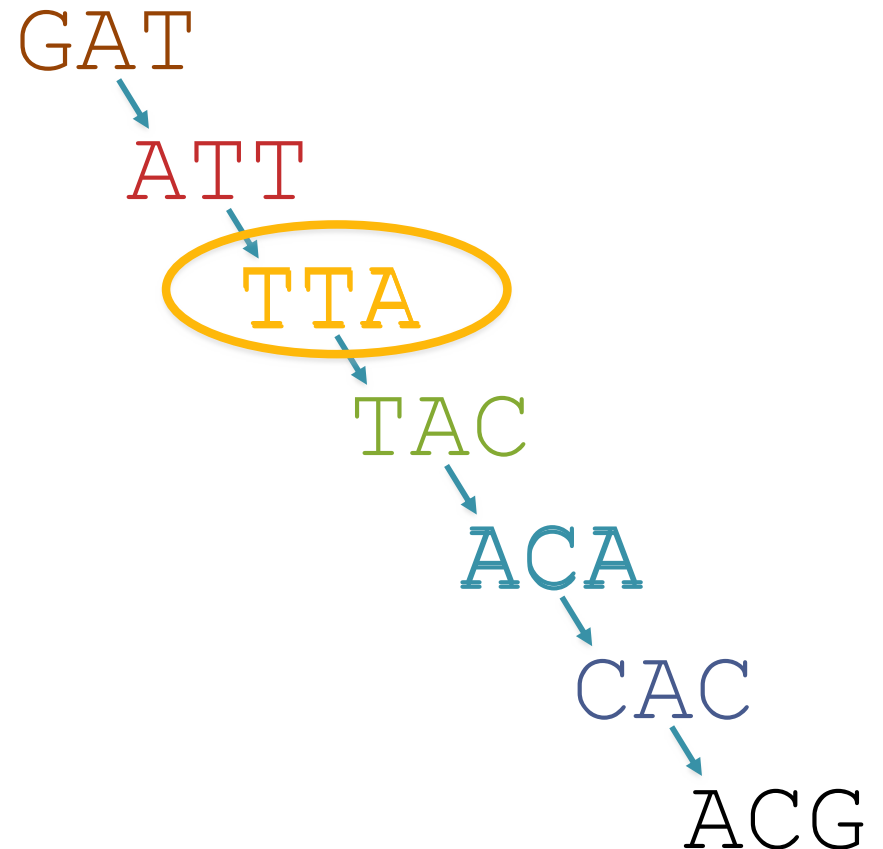
Assemble these reads using a de Bruijn graph approach (k=3):

ACACG

ATTAC

GATTA

TTACA



Pop Quiz I

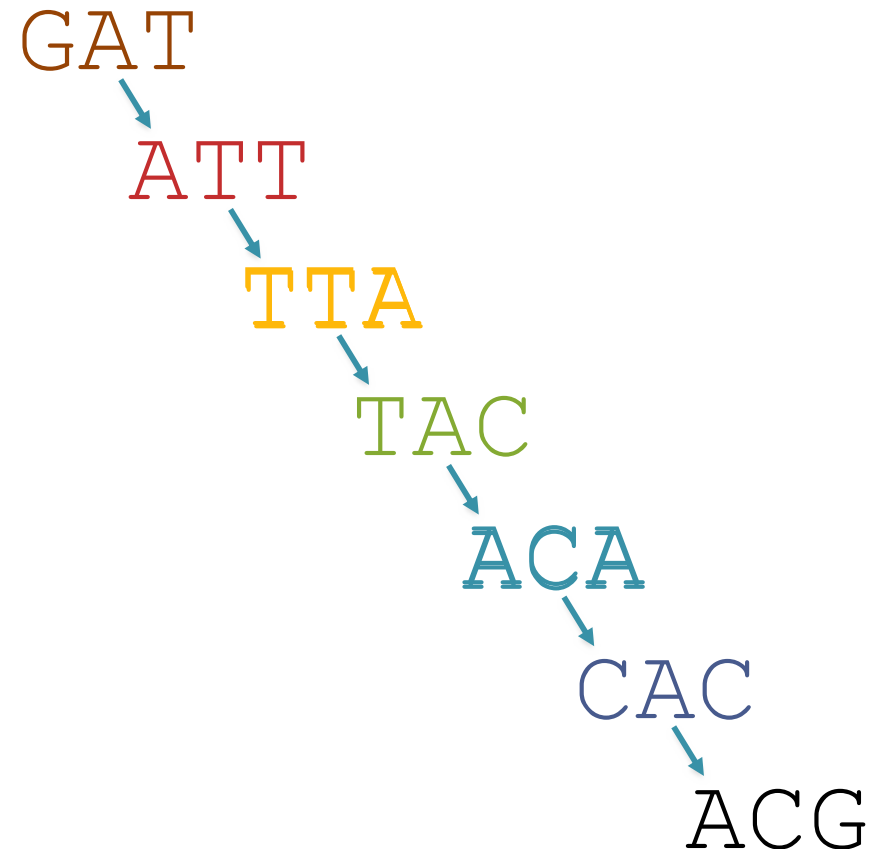
Assemble these reads using a de Bruijn graph approach (k=3):

ACACG

ATTAC

GATTA

TTACA



Pop Quiz I

Assemble these reads using a de Bruijn graph approach (k=3):

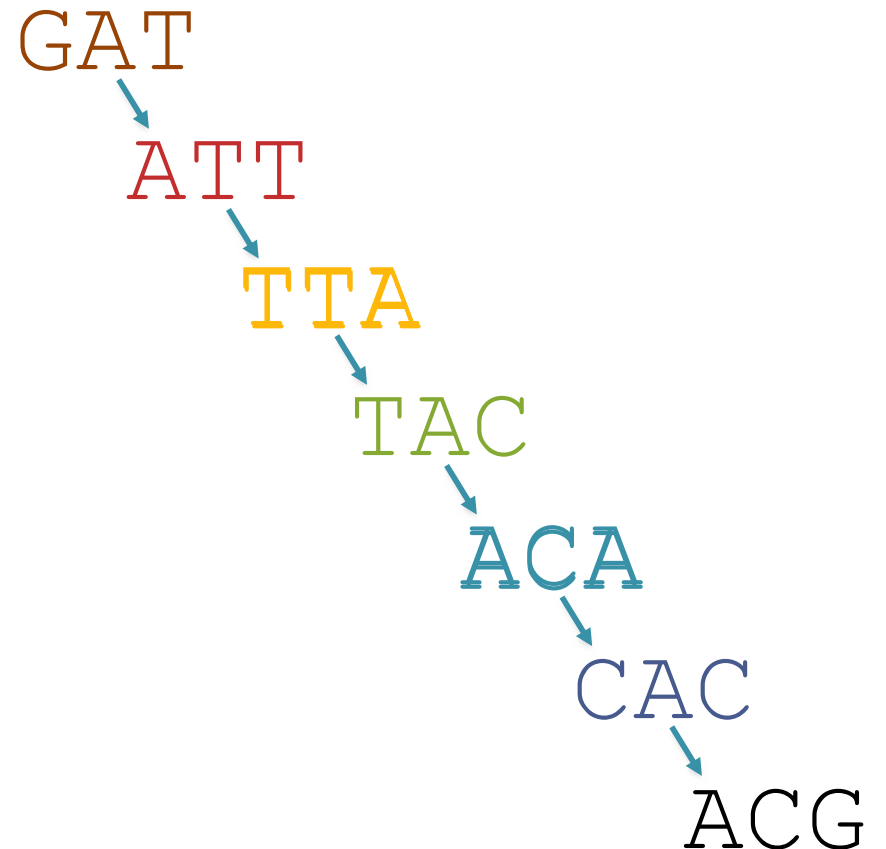
ACACG

ATTAC

GATTA

TTACA

GATTACACG



Pop Quiz I

Assemble these reads using a de Bruijn graph approach (k=3):

ACACG

ATTAC

GATTA

TTACA

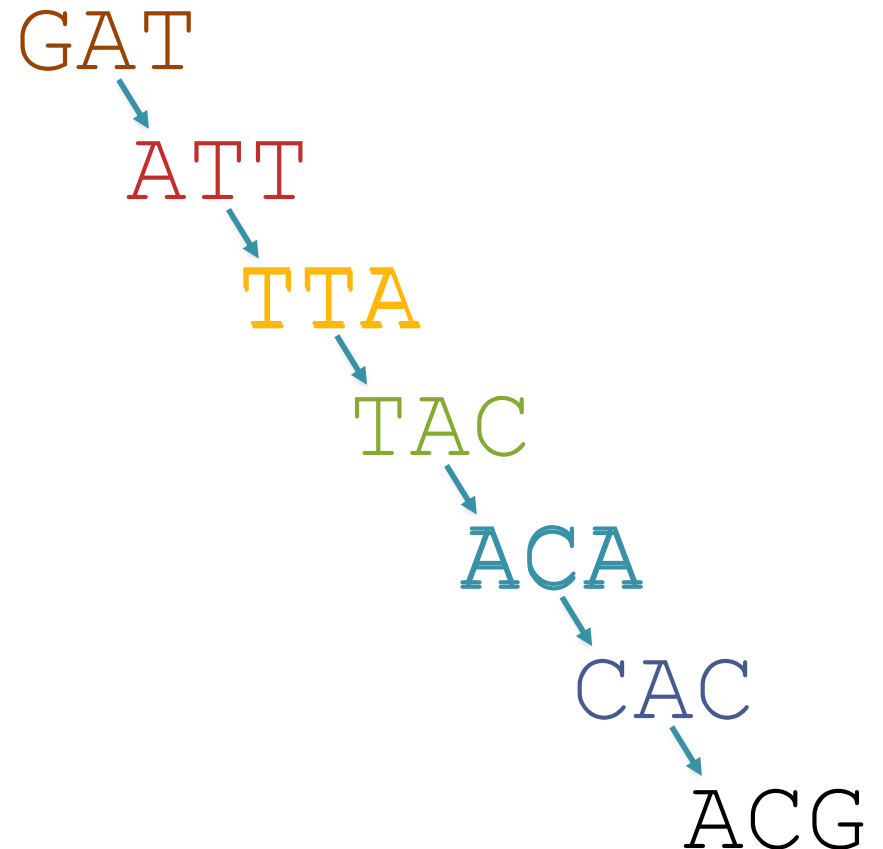
GATTACACG

GATTA

ATTAC

TTACA

ACACG



Pop Quiz 2

Assemble these reads using a de Bruijn graph approach ($k=3$):

ACGA

ACGT

ATAC

CGAC

CGTA

GACG

GTAT

TACG

Pop Quiz 2

Assemble these reads using a de Bruijn graph approach (k=3):

~~ACGA~~

ACGT

ATAC

CGAC

CGTA

GACG

GTAT

TACG

ACG → CGA

Pop Quiz 2

Assemble these reads using a de Bruijn graph approach (k=3):

~~ACGA~~

~~ACGT~~

ATAC

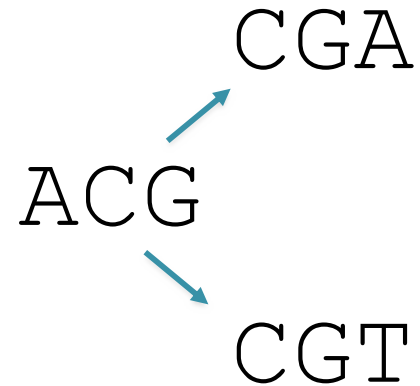
CGAC

CGTA

GACG

GTAT

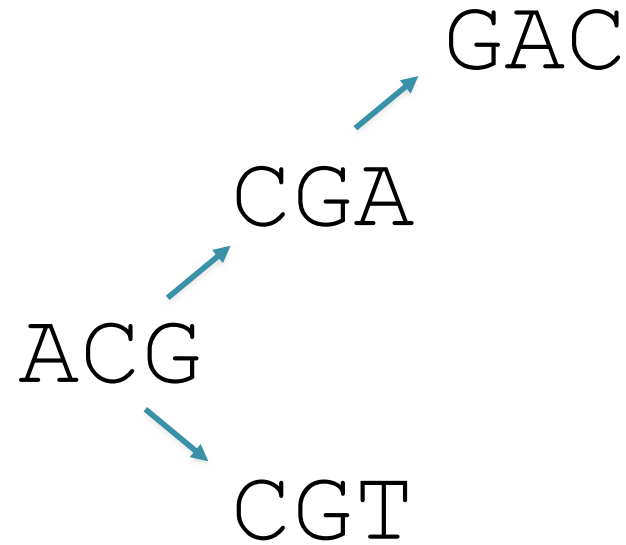
TACG



Pop Quiz 2

Assemble these reads using a de Bruijn graph approach (k=3):

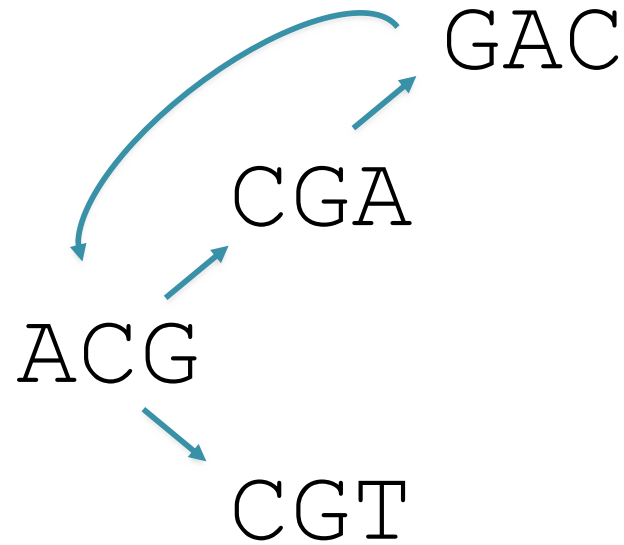
~~ACGA~~
~~ACGT~~
ATAC
~~CGAC~~
CGTA
GACG
GTAT
TACG



Pop Quiz 2

Assemble these reads using a de Bruijn graph approach (k=3):

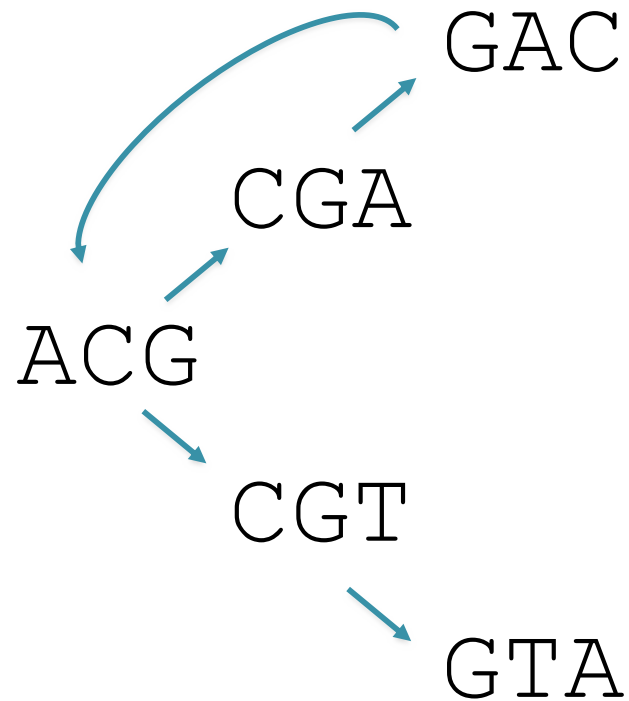
~~ACGA~~
~~ACGT~~
ATAC
~~CGAC~~
CGTA
~~GACG~~
GTAT
TACG



Pop Quiz 2

Assemble these reads using a de Bruijn graph approach (k=3):

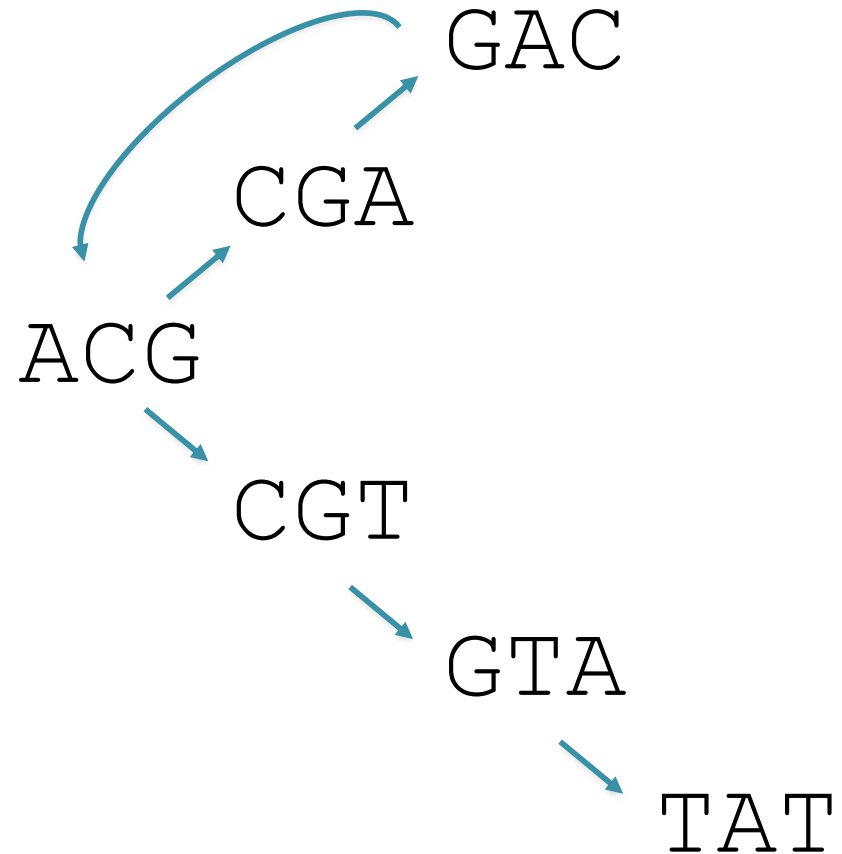
~~ACGA~~
~~ACGT~~
ATAC
~~CGAC~~
~~CGTA~~
~~GACG~~
GTAT
TACG



Pop Quiz 2

Assemble these reads using a de Bruijn graph approach (k=3):

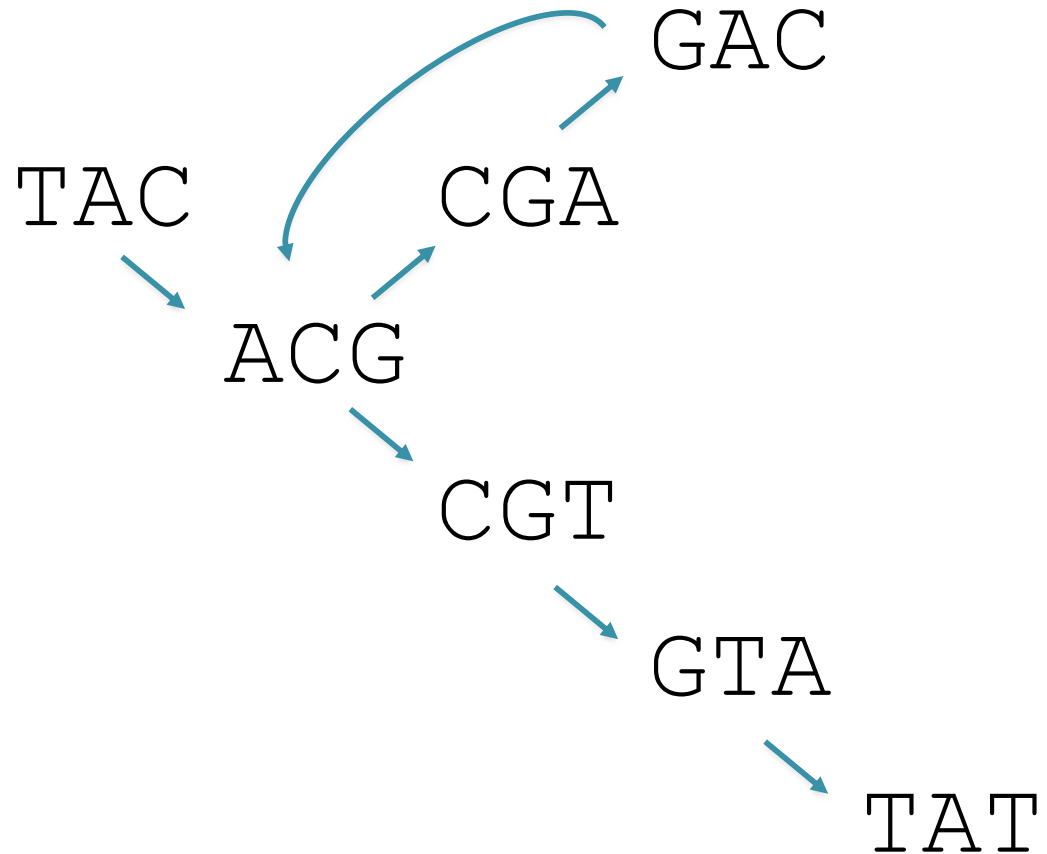
~~ACGA~~
~~ACGT~~
ATAC
~~CGAC~~
~~CGTA~~
~~GACG~~
~~GTAT~~
TACG



Pop Quiz 2

Assemble these reads using a de Bruijn graph approach (k=3):

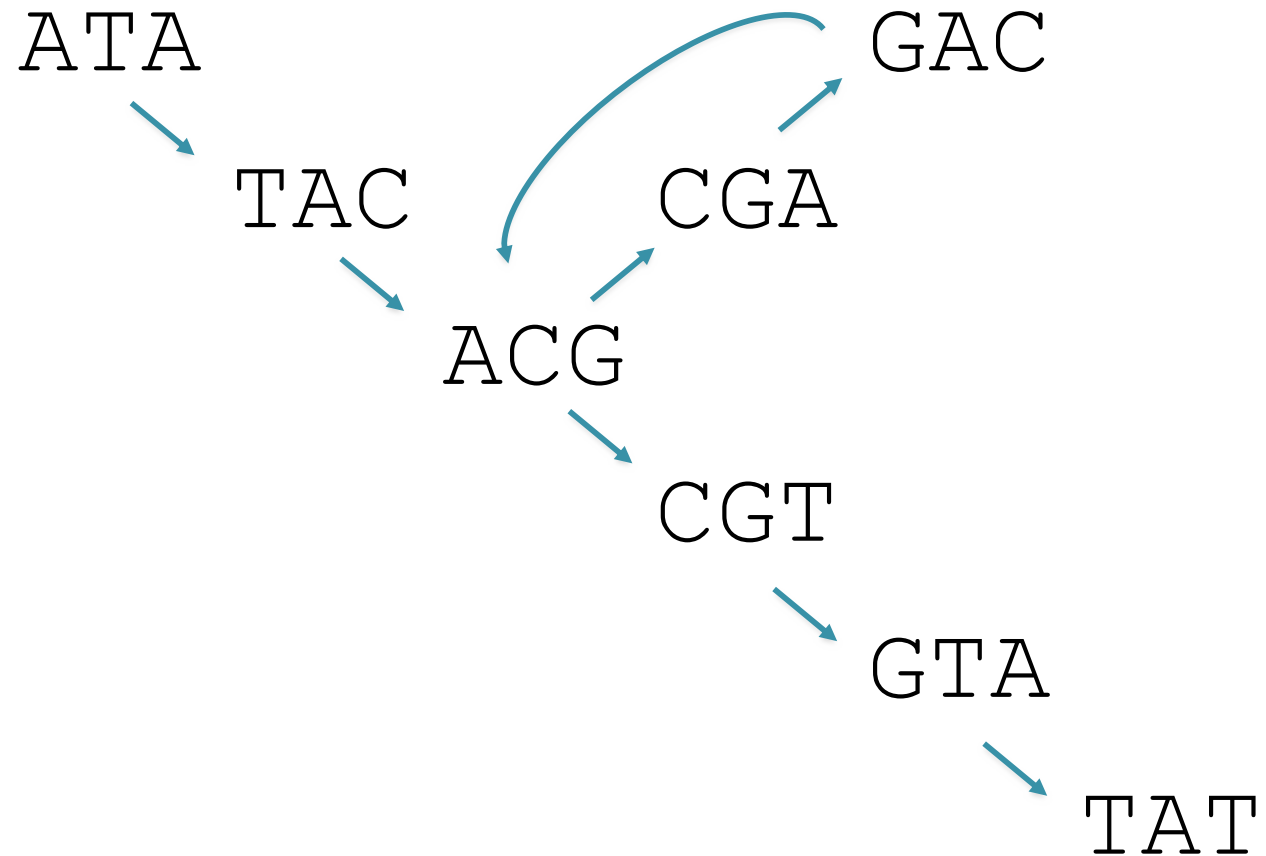
~~ACGA~~
~~ACGT~~
ATAC
~~CGAC~~
~~CGTA~~
~~GACG~~
~~GTAT~~
~~TACG~~



Pop Quiz 2

Assemble these reads using a de Bruijn graph approach (k=3):

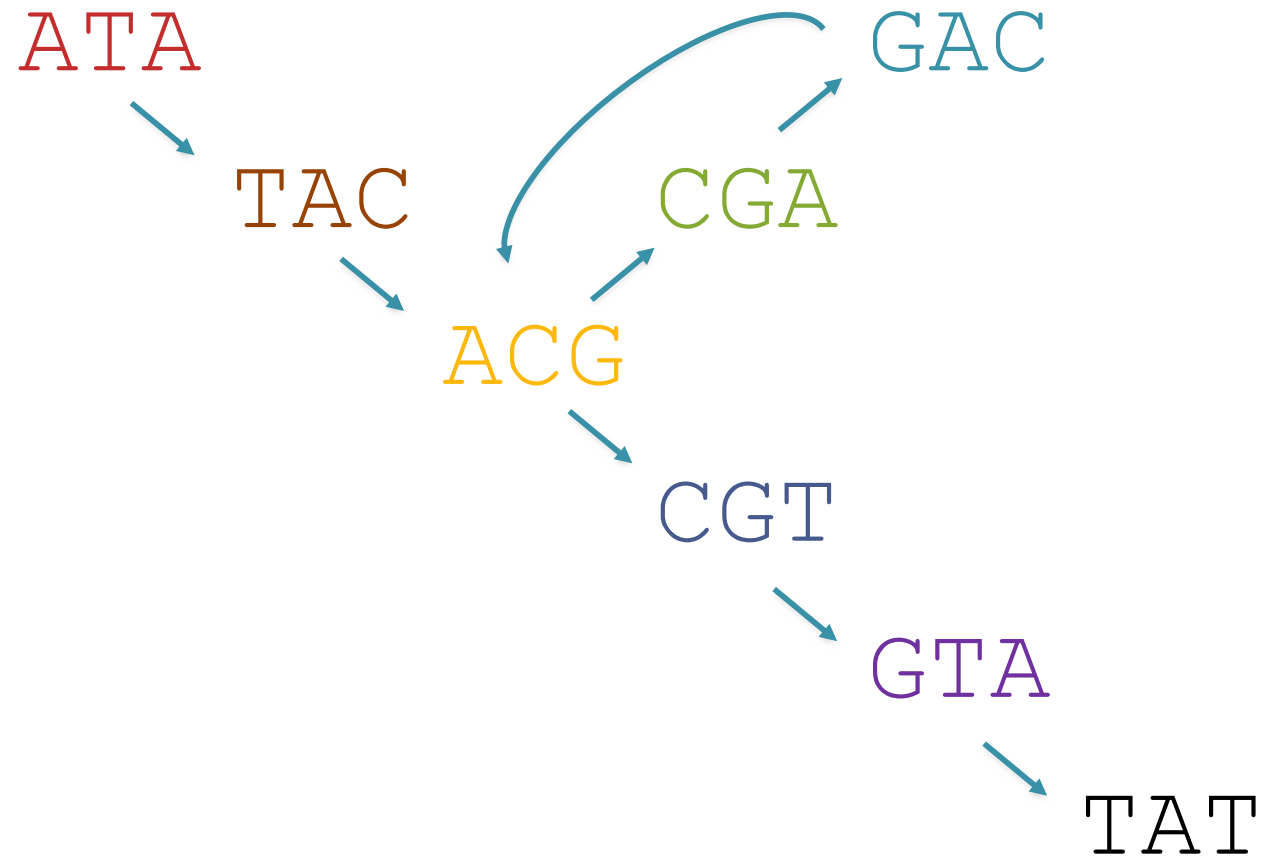
~~ACGA~~
~~ACGT~~
~~ATAC~~
~~CGAC~~
~~CGTA~~
~~GACG~~
~~GTAT~~
~~TACG~~



Pop Quiz 2

Assemble these reads using a de Bruijn graph approach (k=3):

~~ACGA~~
~~ACGT~~
~~ATAC~~
~~CGAC~~
~~CGTA~~
~~GACG~~
~~GTAT~~
~~TACG~~

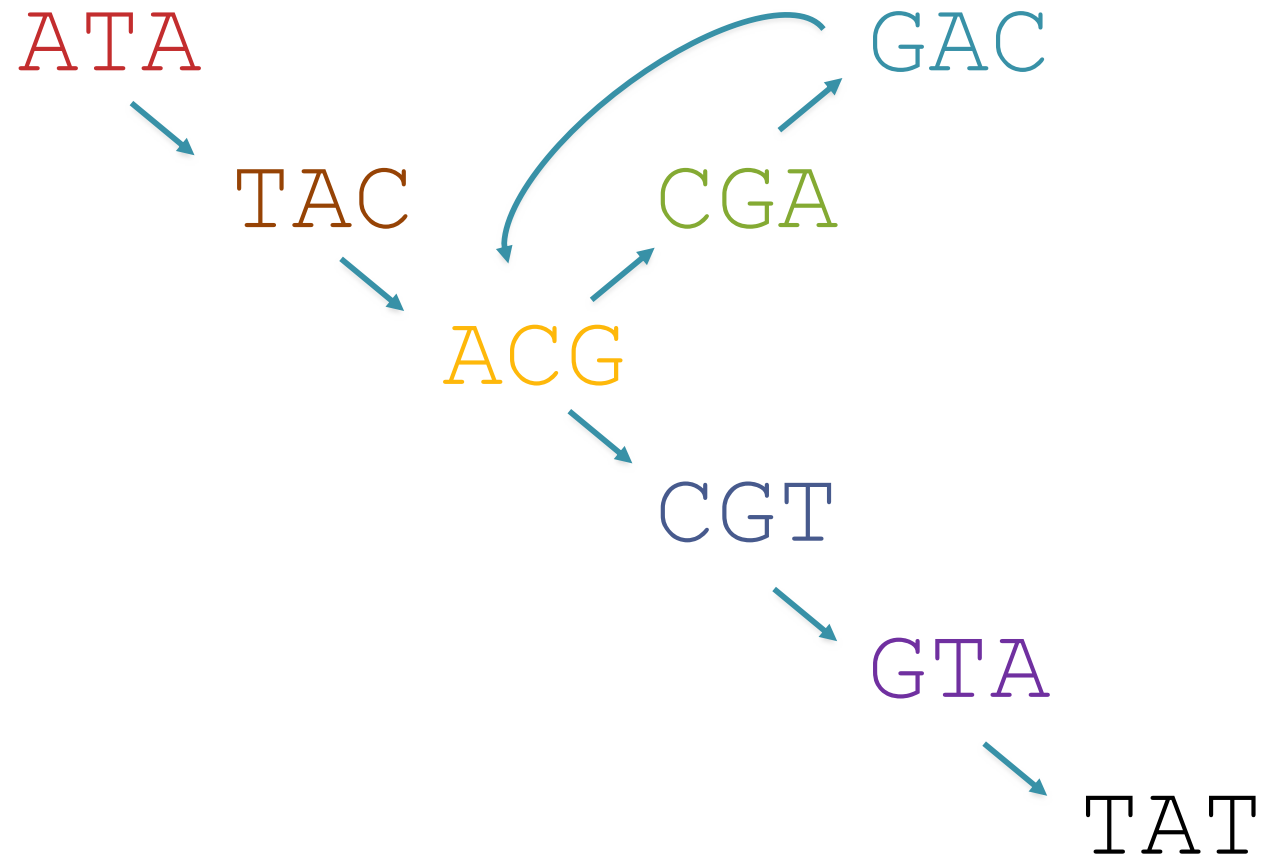


ATACGACGTAT

Pop Quiz 2

Assemble these reads using a de Bruijn graph approach (k=3):

~~ACGA~~
~~ACGT~~
~~ATAC~~
~~CGAC~~
~~CGTA~~
~~GACG~~
~~GTAT~~
~~TACG~~



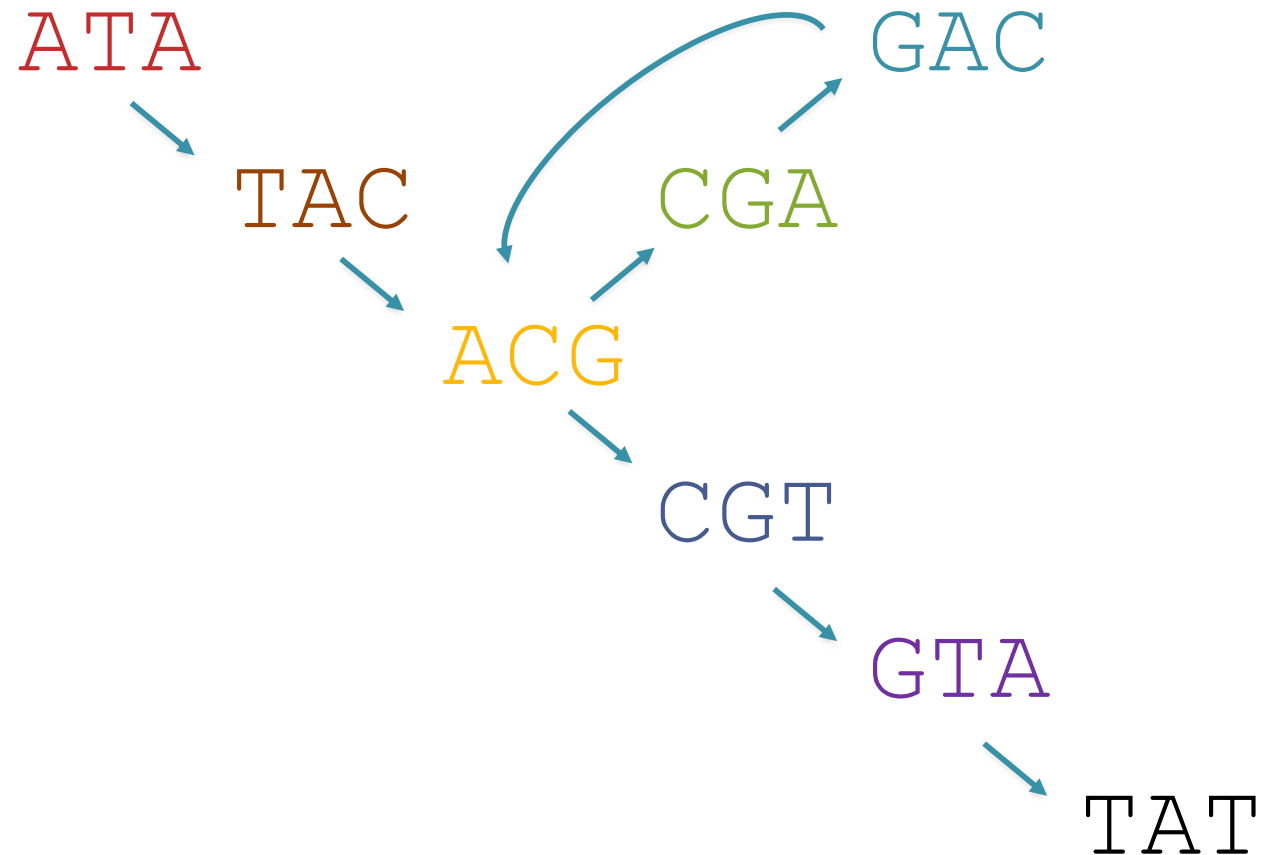
Whats another possible genome?

ATACGACGTAT

Pop Quiz 2

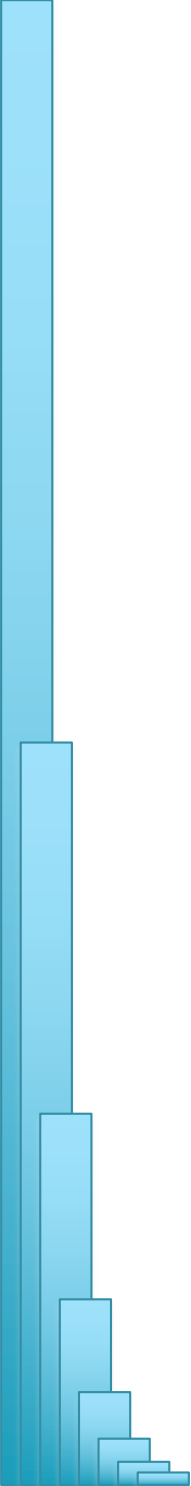
Assemble these reads using a de Bruijn graph approach (k=3):

~~ACGA~~
~~ACGT~~
~~ATAC~~
~~CGAC~~
~~CGTA~~
~~GACG~~
~~GTAT~~
~~TACG~~



Should we add the edge TAT -> ATA?

ATACGACGTAT



Part I. Genome Assembly in Practice

Assembly Applications

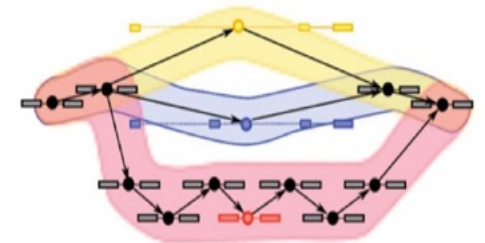
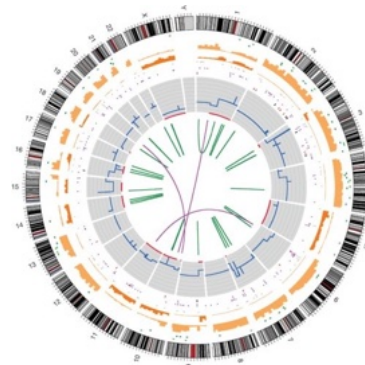
- Novel genomes



- Metagenomes

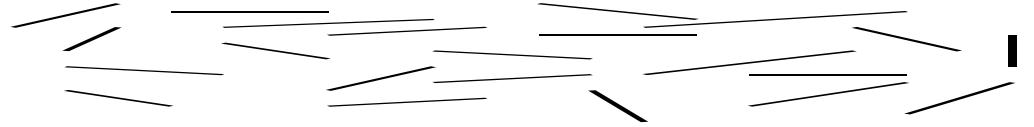


- Sequencing assays
 - Structural variations
 - Transcript assembly
 - ...



Assembling a Genome

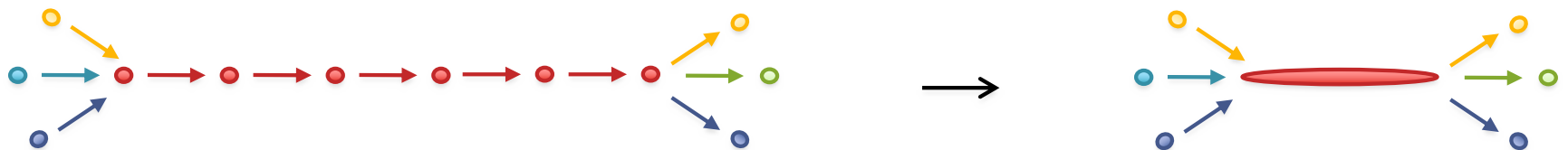
1. Shear & Sequence DNA



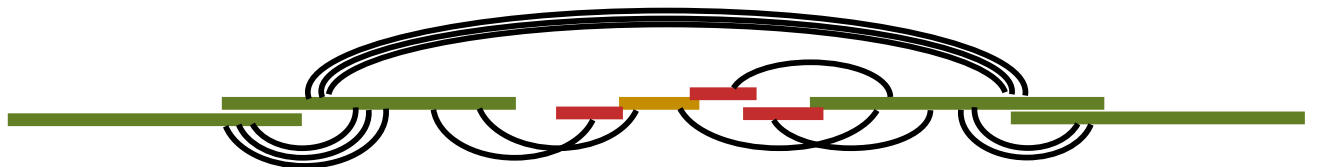
2. Construct assembly graph from reads (de Bruijn / overlap graph)

...AGCCTAGGGATGCGCGACACGT
GGATGCGCGACACGT CGCATATCCGGTTTGGT CAACCTCGGACGGAC
CAACCTCGGACGGACCTCAGCGAA...

3. Simplify assembly graph

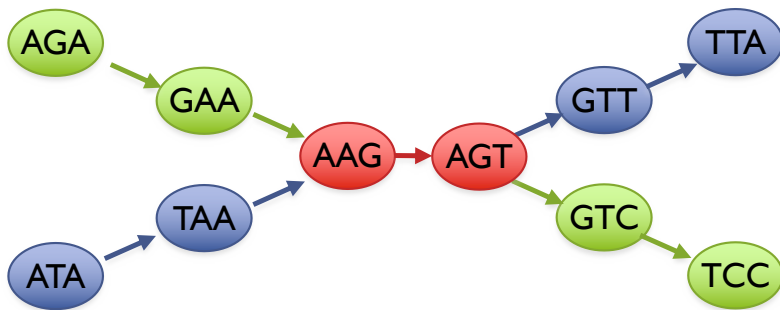


4. Detangle graph with long reads, mates, and other links



Two Paradigms for Assembly

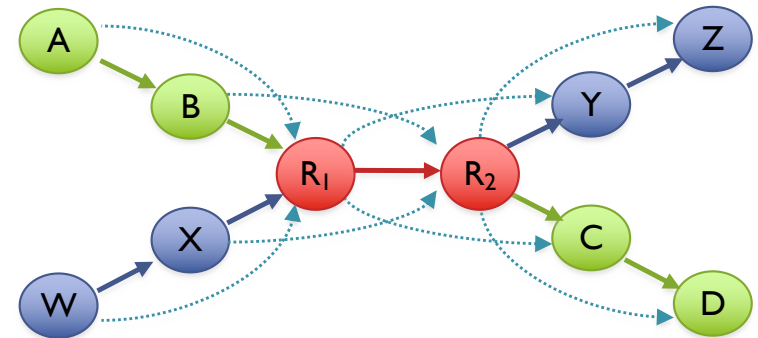
de Bruijn Graph



Short read assemblers

- Repeats depends on word length
- Read coherency, placements lost
- Robust to high coverage

Overlap Graph



Long read assemblers

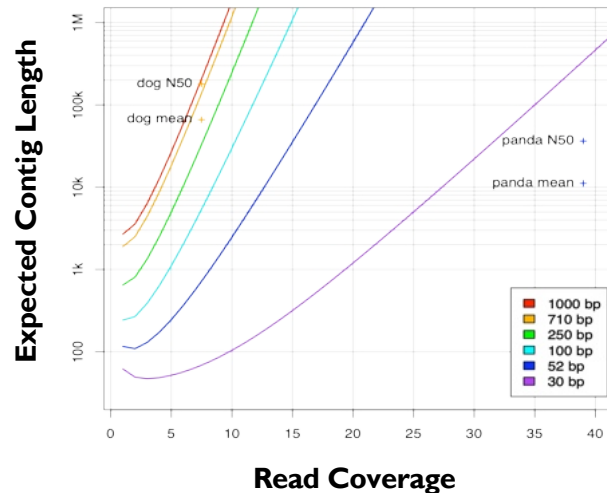
- Repeats depends on read length
- Read coherency, placements kept
- Tangled by high coverage

Assembly of Large Genomes using Second Generation Sequencing

Schatz MC, Delcher AL, Salzberg SL (2010) *Genome Research*. 20:1165-1173.

Ingredients for a good assembly

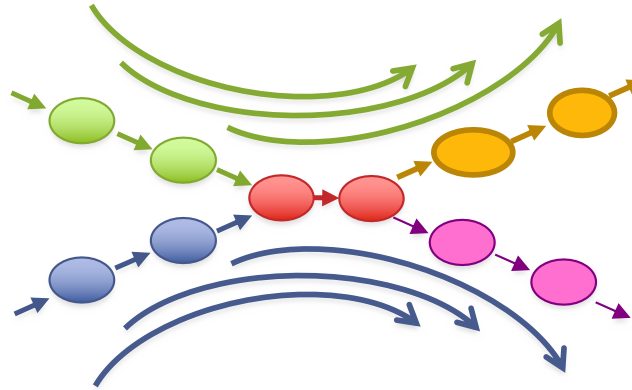
Coverage



High coverage is required

- Oversample the genome to ensure every base is sequenced with long overlaps between reads
- Biased coverage will also fragment assembly

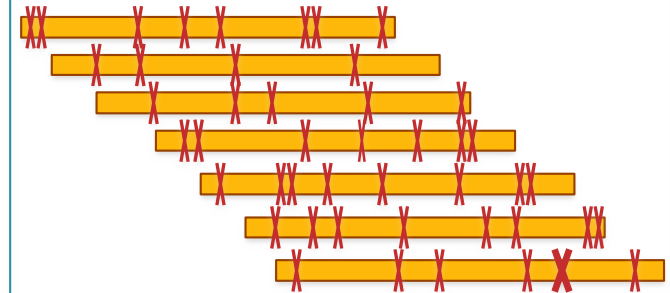
Read Length



Reads & mates must be longer than the repeats

- Short reads will have **false overlaps** forming hairball assembly graphs
- With long enough reads, assemble entire chromosomes into contigs

Quality



Errors obscure overlaps

- Reads are assembled by finding kmers shared in pair of reads
- High error rate requires very short seeds, increasing complexity and forming assembly hairballs

Current challenges in *de novo* plant genome sequencing and assembly

Schatz MC, Witkowski, McCombie, WR (2012) *Genome Biology*. 12:243

Repetitive regions

Repeat Type	Definition / Example	Prevalence
Low-complexity DNA / Microsatellites	$(b_1b_2\dots b_k)^N$ where $1 \leq k \leq 6$ CACACACACACACACACA	2%
SINEs (Short Interspersed Nuclear Elements)	<i>Alu</i> sequence (~280 bp) Mariner elements (~80 bp)	13%
LINEs (Long Interspersed Nuclear Elements)	~500 – 5,000 bp	21%
LTR (long terminal repeat) retrotransposons	Ty1-copia, Ty3-gypsy, Pao-BEL (~100 – 5,000 bp)	8%
Other DNA transposons		3%
Gene families & segmental duplications		4%

- Over 50% of mammalian genomes are repetitive
 - Large plant genomes tend to be even worse
 - Wheat: 16 Gbp; Pine: 24 Gbp

Paired-end and Mate-pairs

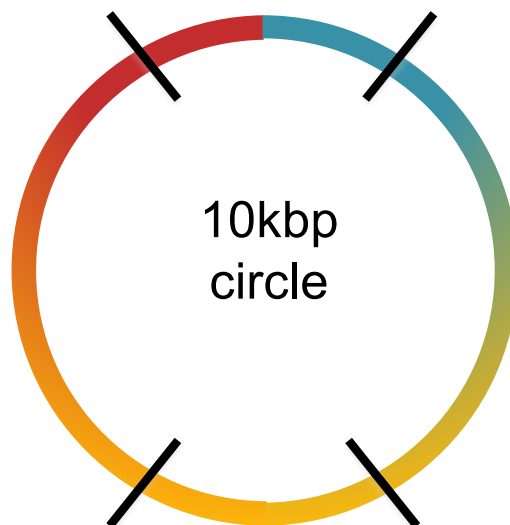
Paired-end sequencing

- Read one end of the molecule, flip, and read the other end
- Generate pair of reads separated by up to 500bp with inward orientation



Mate-pair sequencing

- Circularize long molecules (1-10kbp), shear into fragments, & sequence
- Mate failures create short paired-end reads



2x100 @ ~10kbp (outies)

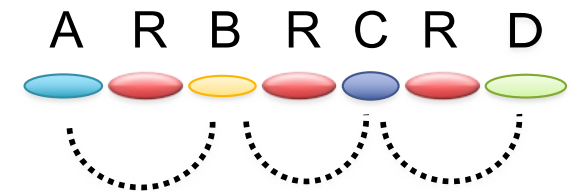
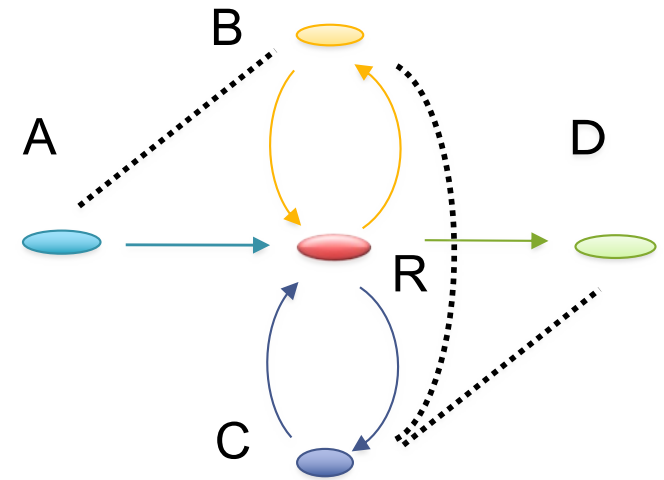


2x100 @ 300bp (innies)



Scaffolding

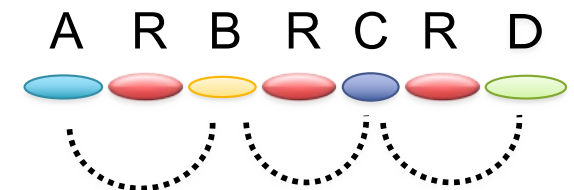
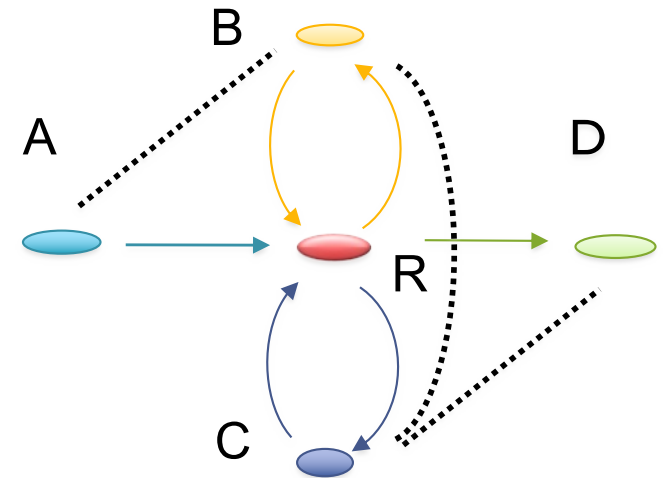
- Initial contigs (*aka* unipaths, unitigs) terminate at
 - *Coverage gaps*: especially extreme GC
 - *Conflicts*: errors, repeat boundaries
- Use mate-pairs to resolve correct order through assembly graph
 - Place sequence to satisfy the mate constraints
 - Mates through repeat nodes are tangled
- Final scaffold may have internal gaps called sequencing gaps
 - We know the order, orientation, and spacing, but just not the bases. Fill with Ns instead



Why do contigs & scaffolds end?

Scaffolding

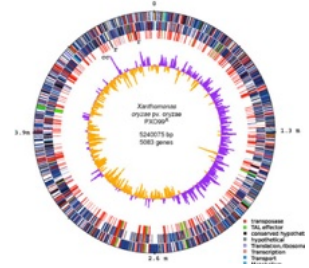
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Why do contigs & scaffolds end?

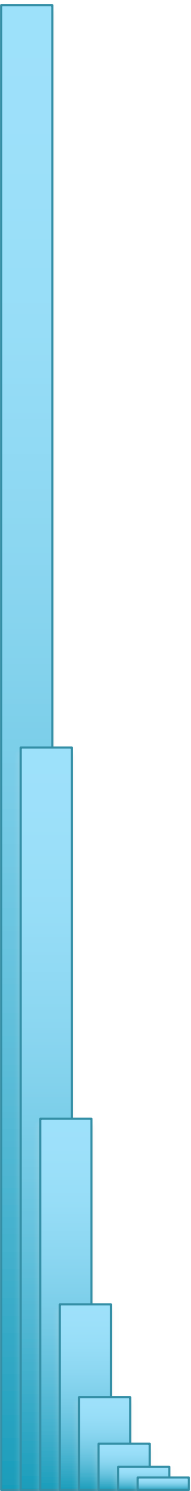
(1) end of chromosome; (2) gap in coverage; (3) repeats; (4) heterozygosity

Assembly Summary



Assembly quality depends on

1. **Coverage**: low coverage is mathematically hopeless
 2. **Repeat composition**: high repeat content is challenging
 3. **Read length**: longer reads help resolve repeats
 4. **Error rate**: errors reduce coverage, obscure true overlaps
- Assembly is a hierarchical, starting from individual reads, build high confidence contigs/unitigs, incorporate the mates to build scaffolds
 - Extensive error correction is the key to getting the best assembly possible from a given data set
 - Recommend spades for short read assembly
 - Integrates error correction and scaffolding



Part II. The human genome

The scale of DNA in our body is staggering.

- A typical human is comprised of **roughly 40 trillion human cells**
(excluding trillions of bacterial cells in our gut)
- If stretched out, each haploid genome would be **roughly 2 meters**.
- So, each cell has 4 meters of DNA.
- $40 \text{ trillion} * 4 \text{ meters} = 160 \text{ trillion meters}$.
- $160 \text{ trillion meters} / 1609.34 = 99,750,623,441 \text{ miles}$
- $99,750,623,441 / 92,960,000 = 1,073.05 \text{ trips to the sun}$.

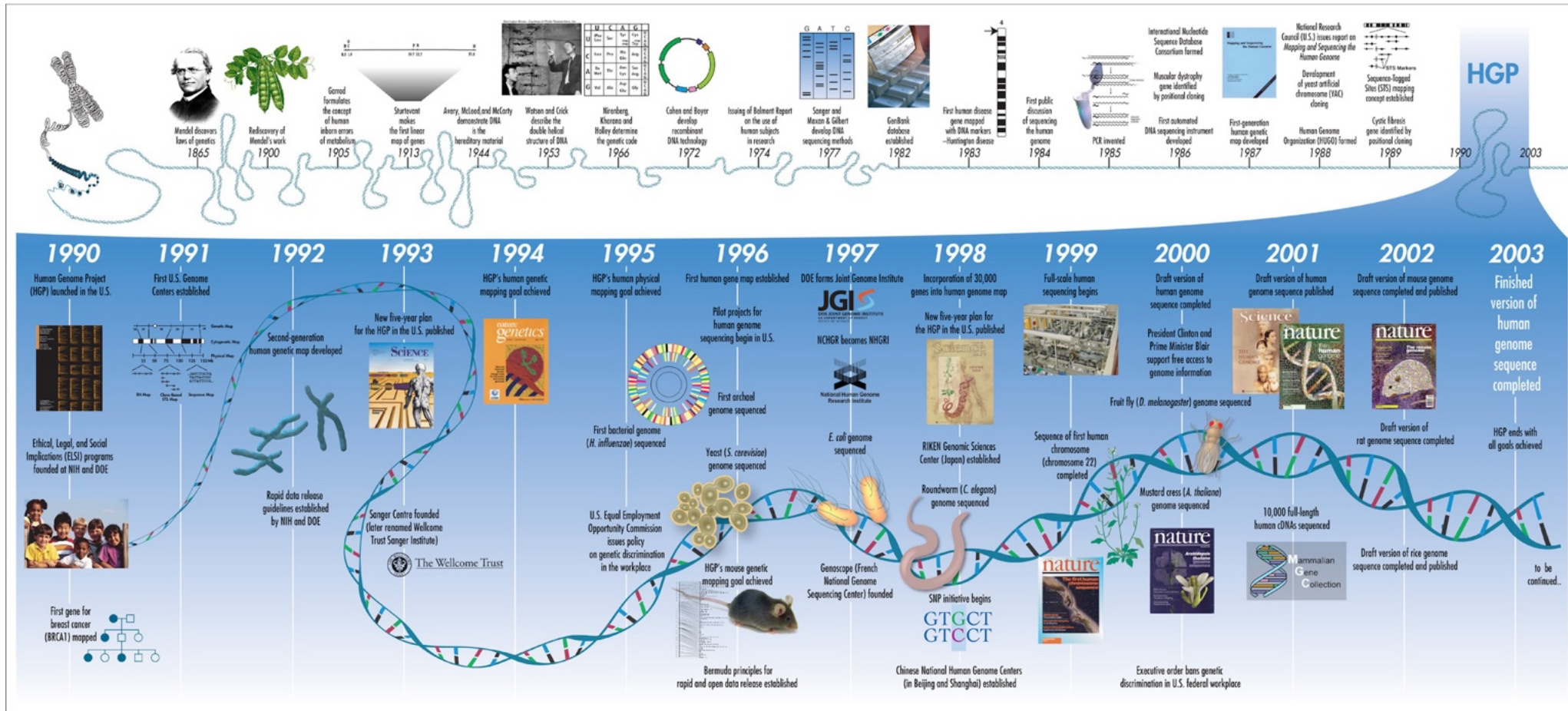
A typical cell replicates about 100 times

160 trillion meters x 100 =

1.69123746 light years

[More info](#)

History of the Human Genome Project



The reference human genome



“Without a doubt, this is the most important, most wondrous map ever produced by humankind.”

*Bill Clinton
June 26, 2000*

The reference human genome



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The Sequence of the Human Genome

Venter et al.

Science 291, pp 1304-1351 (2001)

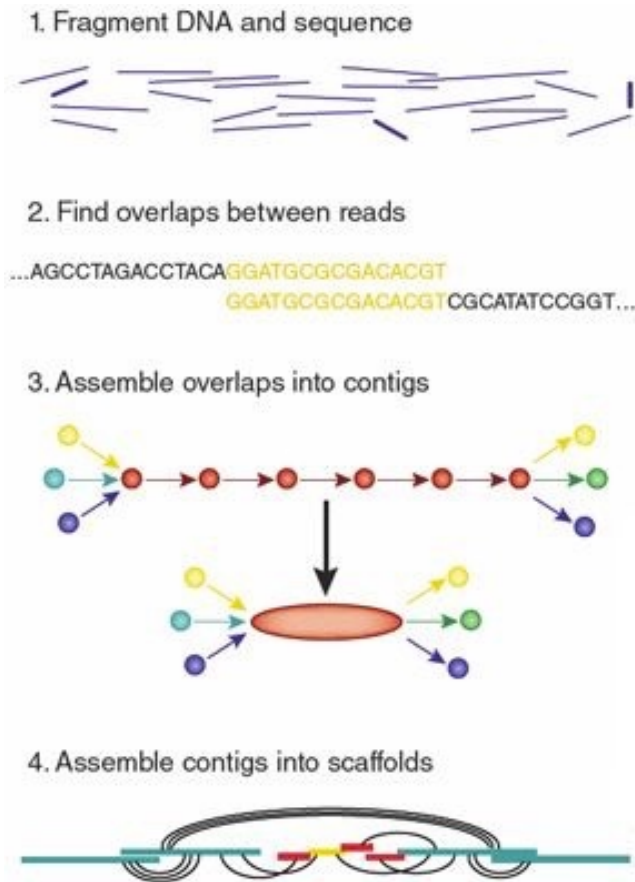


Initial sequencing and analysis of the human genome

International Human Genome Sequencing Consortium

Nature 409, pp 860-921 (2001)

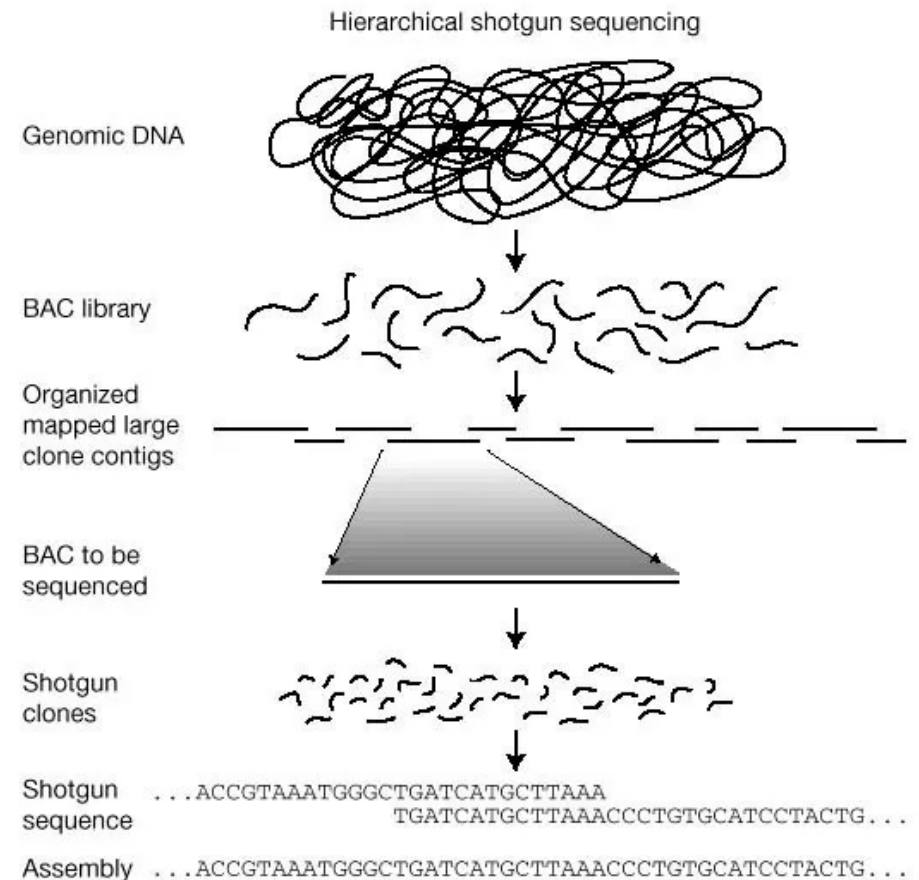
Two Human Genomes?



The Sequence of the Human Genome

Venter et al.
Science 291, pp 1304-1351 (2001)

(Figure from Baker (2012) Nature Methods)



Initial sequencing and analysis of the human genome

International Human Genome Sequencing Consortium
Nature 409, pp 860-921 (2001)

Who is the reference human?

The Buffalo News/Sunday, March 23, 1997

Government abuse, civil disobedience

people. But the very nature of government creates a mind set that inspires to increase their authority, always at the expense of the people."

ly, government has forgotten that servant of the people," Parlato adds, "acting more like it's the master." to and the Lapps share an abiding non-violent civil disobedience. insist on being respectful in our of resistance," Barbara Lyn Lapp but if we claim to care about our rs, we must protest government in-

violence has to be the watchword, said, calling civil disobedience the is of the violent militia movement. on-violence can serve as an anti-government oppression, he added. law is unjust or you're given an without moral or legal authority,

you should refuse it," Parlato said. "And, if need be, you have to be brave enough to accept the consequences."

Rachel Lapp says she believes government can be good, when it controls the aggressors in society. Instead, it too often comes down on the side of the aggressors, who enforce child-protection laws, compulsory education, disclosure rules on tax forms and seat belt laws.

"We want people to see the correlation between what happened to us and what can happen to anyone when government gets out of hand," Rachel Lapp said.

The Lapps and Parlato will be joined by Samuel Radford III, a critic of public education who was arrested and pleaded guilty to reduced charges following a 1993 disturbance at the City Campus of Erie Community College.

WANTED

20 Volunteers

to participate in the

Human Genome Project

a very large international scientific research effort.

The goal is to decode the human hereditary information (*human blueprint*) that determines all individual traits inherited from parents. The outcome of the project will have tremendous impact on future progress of medical science and lead to improved diagnosis and treatment of hereditary diseases.

Volunteers will receive information about the project from the Clinical Genetics Service at Roswell Park, and sign a consent form before participating.

No personal information will be maintained or transferred.

Volunteers will provide a one-time donation of a small blood specimen. A small monetary reimbursement will be provided to the participants for their time and effort.

Individuals must be at least 18 years of age.
Persons who have undergone chemotherapy are not eligible.

For more information please contact the
Clinical Genetics Service
845-5720 (9:00 am - 3:00 pm)
March 24 - 26, 1997

ROSWEIL PARK
CANCER INSTITUTE

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Pieter de Jong, RPCI

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to and the Lapps share an abiding commitment to non-violent civil disobedience. "We insist on being respectful in our acts of resistance," Barbara Lyn Lapp said, "but if we claim to care about our children, we must protest government involvement in violence has to be the watchword," she said, calling civil disobedience the only alternative to the violent militia movement. "Non-violence can serve as an antidote to government oppression," he added. "If the law is unjust or you're given an option without moral or legal authority, you should refuse it," Parlato said. "And, if need be, you have to be brave enough to accept the consequences."

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March 24 - 26, 1997

Appendix: Identifying the ancestry of segments of the human genome reference sequence

To compare Neandertal to present-day human haplotypes for the purpose of population genetic analysis, we needed to have long haploid sequences from present-day humans that were of known ancestry. To identify such segments, we took advantage of the fact that the human reference sequence is haploid over scales of tens of kilobases, because it is comprised of a tiling-path of Bacterial Artificial Chromosomes (BACs) or other clone types that are of typical size 50-150 kb (S92). We do not know of any other substantial source of high quality human haploid sequences of the requisite size.

Determining the ancestries of the libraries in the human genome reference sequence using HAPMIX

It is crucial to know the 'ancestry' of a clone to use it in a meaningful population genetic analysis. In what follows, we define 'ancestry' as the geographic region in which a clone's ancestor lived 1,000 years ago, inferred based on its genetic proximity to other individuals from that region today. This definition allows us to classify clones from Chinese Americans as "East Asian," from European Americans as "European", and from African Americans as either "West African" or "European".

To identify the ancestries of the libraries comprising most of the human genome reference sequence, we used a list of 26,558 clones tiling the great majority of the genome, most of which we were able to assign to a library of origin. Restricting to the autosomes, we identified 21,156 clones that seemed to fall into 9 libraries based on the naming scheme: CTA (n=199), CTB (n=356), CTC (n=452), CTD (n=1,426), RPCI-1 (n=740), RPCI-3 (n=456), RPCI-4 (n=716), RPCI-5 (n=802) and RPCI-11 (n=16,009). (In a subsequent re-examination, we identified additional clones that we likely could have classified into libraries, including 953 from RPCI-11, 632 from RPCI-1, and 490 from another library RPCI-13.) The median span of the 21,156 clones we analyzed was 112 kb, and 80% are >50kb in size. About 2/3 came from a single library, RPCI-11.

1. **RPCI-11 is an African American:** RPCI-11, the individual who contributed most of the human genome reference sequence, is consistent with having African American ancestry, with 42% of the clones of confident West African ancestry and 42% of the clones of confident European ancestry, and the ancestry of the remaining clones less confidently inferred. The finding of likely African American ancestry for RPCI-11 was previously reported in a study of the ancestry of RPCI-11 clones spanning the Duffy blood group locus (S93), and here we confirm this finding, and also expand the inference to the whole genome.
2. **CTD is an East Asian:** The majority of clones from CTD, the second largest library in its contribution to the human genome sequence, is likely an East Asian. In a HAPMIX analysis with CEU (European) - CHB+JPT (East Asian) as the proposed ancestral populations, the majority of clones are of confident East Asian origin, and there is no secondary mode of confident European ancestry, as might be expected from a Latino or South Asian individual.
3. **The remaining 7 libraries are European:** The remaining libraries (CTA, CTB, CTC, RPCI-1, RPCI-3, RPCI-4 and RPCI-5) are inferred to be of European ancestry, since they all have consistent distributions of inferred clone ancestries, with the majority of clones of confident European ancestry in both our HAPMIX analyses and no secondary modes.

A Draft Sequence of the Neandertal Genome

Green et al (2010) Science. DOI: 10.1126/science.1188021

Supplemental Note 16 (pg 145-146)

Pieter de Jong, RPCI

Who is the reference human?

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EDITORIAL
Nature Methods **7**, 331 (2010)
[doi:10.1038/nmeth0510-331](#)
E pluribus unum
If the human reference genome is to reflect more of the actual genomic diversity in humans, community participation is needed.
[Please visit methagora to view and post comments on this article.](#)

The human genome is ten years old. We acknowledge its reference assembly as an invaluable resource essential for many purposes such as the assembly of short reads from high-throughput sequencing platforms into chromosome context during resequencing projects. At the same time, we think necessary improvement of the reference genome depends on the willingness of the research community to provide data for the genome's less accessible regions.

First published in 2001, the human reference genome has, since 2007, been in the hands of the Genome Reference Consortium (GRC) a small group of fewer than 20 scientists from the European Bioinformatics Institute, the US National Center for Biotechnology Information, The Sanger Institute and The Genome Center at Washington University in St. Louis, who have committed to the improvement and completion of this reference, with very little financial support.

The reference genome is now in its 19th rendition, and probably the best measure of its improvement over the last ten years is the number of fragments it consists of. The very first version had ~150,000 gaps; the most recent build, GRCh37, has only around 250 gaps.

The only other publicly accessible *de novo* assembly of a human genome that contains chromosome sequences is HuRef. Obtained by traditional capillary sequencing, HuRef is the diploid genome of Craig Venter. It comes in 4,500 pieces and, like any individual genome, it contains many rare alleles.

GRCh37, in contrast, is a mosaic haploid genome derived from about 13 people. It still contains rare alleles, but the GRC recently decided to convert these to common haplotypes. Deciding which alleles are common and which are rare is proving challenging, and the GRC members are collaborating with members of the 1000 Genomes project to collect enough data to make these decisions.

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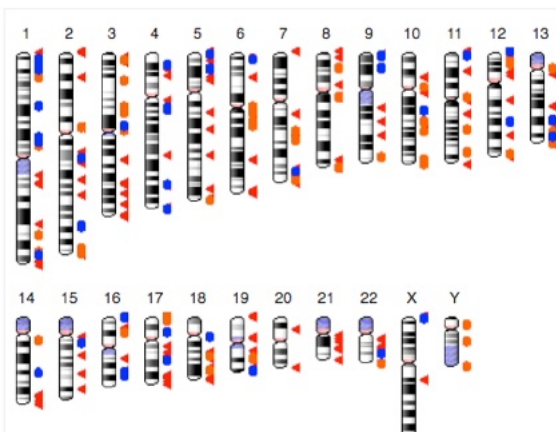
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Human Genome Overview

Information about the continuing improvement of the human genome



- ◀ Region containing alternate loci
- ◀ Region containing fix patches
- ◀ Region containing novel patches

Ideogram of the latest human assembly, GRCh38.p11

The GRC is working hard to provide the best possible by both generating multiple representations (alternates) represented by a single path. Additionally, we are releasing allows users who are interested in a specific locus to affecting users who need chromosome coordinate s

Download data:

- [GRCh38.p11 \(latest minor release\) FTP](#)
- [GRCh38 \(latest major release\) FTP](#)
- [Genomic regions under review FTP](#)
- [Current Tiling Path Files \(TPFs\)](#)

Transitioning to GRCh38? Try the [NCBI Remapping](#) assembly alignments used by the GRC.

Next assembly update

The next assembly update (GRCh38.p12) will be


[GRCh38.p11](#)
[GRCh37.p13](#)
[GRCh37](#)

GRCh38.p11

Release date: June 14, 2017

Release type: minor

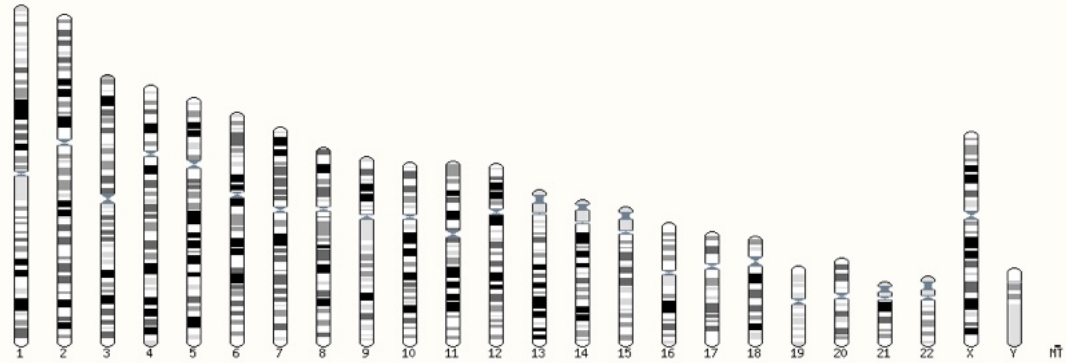
Release notes: GRCh38.p11 is the eleventh patch release for the GRCh38 reference assembly. No chromosome coordinate of patch scaffolds is now: 64 FIX and 59 NOVEL.

Assembly accessions: GenBank: [GCA_000001405.26](#) , RefSeq: [GCF_000001405.37](#)

Pseudoautosomal regions

Name	Chr	Start	Stop
PAR#1	X	10,001	2,781,479
PAR#2	X	155,701,383	156,030,895
PAR#1	Y	10,001	2,781,479
PAR#2	Y	56,887,903	57,217,415

The human genome - basic stats



- 3.096 billion base pairs (haploid)
- 20,454 protein coding genes
- 226,950 coding transcripts
(isoforms of a gene that each encode a distinct protein product)

Assembly	GRCh38.p12 (Genome Reference Consortium Human Build 38), INSDC Assembly GCA_000001405.27 , Dec 2013
Base Pairs	3,609,003,417
Golden Path Length	3,096,649,726
Annotation provider	Ensembl
Annotation method	Full genebuild
Genebuild started	Jan 2014
Genebuild released	Jul 2014
Genebuild last updated/patched	Mar 2019
Database version	97.38
Gencode version	GENCODE 31

Gene counts (Primary assembly)

Coding genes	20,454 (incl 660 readthrough)
Non coding genes	23,940
Small non coding genes	4,871
Long non coding genes	16,848 (incl 302 readthrough)
Misc non coding genes	2,221
Pseudogenes	15,204 (incl 8 readthrough)
Gene transcripts	226,950

SPECIAL REPORT

The untold story of the Human Genome Project: How one man's DNA became a pillar of genetics

By Ashley Smart — Undark July 9, 2024



Pieter De Jong, who led the Human Genome Project work at Roswell Park Cancer Institute, stores DNA samples from the anonymous donor known as RP11 (or RPCI-11) at minus 80 degrees Celsius in a freezer at his home in Redmond, Wash.

JOVELLE TAMAYO FOR STAT



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STAT is co-publishing this [investigation by Undark](#).



hey numbered 20 in all — 10 men and 10 women who came to a sprawling medical campus in downtown Buffalo, N.Y., to volunteer for

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